THE DIGITAL PSYCHE

A PSYCHOLOGICALLY INFORMED RESEARCH AGENDA FOR AI SAFETY AND ETHICS

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ABSTRACT

Some recurrent AI failures appear as patterns across situations rather than defects visible in one output. A failure observed only once may warrant incident analysis, but it does not by itself establish a disposition or candidate mechanism. This Perspective asks when psychology helps explain those patterns without turning metaphor into diagnosis. A pattern must first remain reliable under irrelevant perturbations, predict behavior on an independent task, and change under selective intervention. Those three gates bear on a candidate mechanism. The psychological name faces another test: it earns a role only when it adds value through prediction, causal discrimination, or intervention choice beyond a matched engineering account; otherwise, we keep the engineering result and release the clinical name. The paper applies this rule to automatic perspective sensitivity, aversive learning, and workspace-like integration, where current evidence does not establish a machine disorder or human-like mechanism but does motivate each inquiry across current models and agentic systems. We then turn the psychological lens toward the human–AI interface and develop the Harm-Gated Workspace as a bounded architectural hypothesis. Three comparative protocols make the engineering claims rejectable, and two of them propose the further test a psychological name would need. None has yet been run. Machine Psychology earns its place through experiments, not labels.

Keywords AI safety · machine psychology · behavioral evaluation · psychological analogy · agentic AI · human–AI interaction · theory of mind · cognitive architectures · AI ethics

1 Part I: Introduction. The Ghost in the Machine

1.1 The New Frontier of AI and the Limits of Traditional Safety

Earlier systems mastered bounded tasks such as chess and image recognition, and their edges remained visible. What is changing now is not only performance but scope. Frontier models work across language and code, often through multimodal interfaces, and the pursuit of Artificial General Intelligence (AGI) has moved from philosophical speculation to an engineering objective whose threshold remains contested and whose arrival remains uncertain (Morris et al., 2024). The frontier is shifting again. The AI co-scientist generates and refines research hypotheses, although people remain responsible for experimental validation (Gottweis et al., 2026). The ARC Prize reports progress but limited generalization to novel tasks (Chollet et al., 2024, 2026).

This new frontier carries a special safety problem because learned behavior depends on how a system is trained and where it is used, so established software safety remains indispensable. Engineers still need to test bounded components and control access, then monitor the system after deployment. Formal methods can provide strong guarantees for specified properties, but a complete account examines more than source code alone.

The underlying problem is that some malfunctions appear as recurrent behavior rather than isolated code defects. Models sometimes invent facts or change their conduct across contexts, yet such patterns emerge from model development and conditions of use rather than from any single line of code. Exhaustive testing is unrealistic, and formal verification reaches only specified properties in bounded systems. The ghost in the machine is not an argument against engineering but a problem of level. The engineering task still includes tracing components and controls, yet we also need a language for stable patterns across situations that suggests competing causes and lets us compare interventions.

1.1.1 The Psychopathological Framework

This paper adds concepts from psychology and neuroscience to the engineering toolkit in order to study recurrent behavior at the level where users and institutions encounter it. Its central claim is testable. A psychological analogy earns a role only after evidence supports a candidate mechanism and the name adds value beyond a matched engineering account. Section 3.1.1 condenses that test into a practical checklist before the full protocols.

We do not infer that an AI suffers as a person does; the narrower claim is that stable maladaptive behavior may support functionally analogous hypotheses about information processing. The same output can arise from several causes, and output alone cannot identify an architecture. With that boundary in place, the framework organizes how we assess recurrent behavior and mitigate it, and uses what we learn to inform safer design. Diagnosis and treatment remain bounded research metaphors because the framework cannot guarantee alignment; its purpose is to sharpen measurement and force competing interventions into view.

1.1.2 Roadmap of the Paper

The paper presents a systematic case for using psychopathology to generate hypotheses in AI safety, moving from the human analogy to machine assays before asking what the framework changes for people and for the design of systems.

Part II preserves disagreement within psychopathy research and builds the human analogue around two candidate dimensions, automatic sensitivity to another perspective and learning from aversive outcomes. They neither exhaust psychopathy nor define a settled human architecture, and they are not criteria for machines. The section then treats empathy, prosocial action, and information integration as distinct capacities.

Part III crosses from the human analogue to the artificial patient by examining unstable psychometric output, explicit social reasoning, and broader behavior after narrow fine-tuning. Those findings expose patterns worth explaining rather than evidence of a fragmented self or hidden personality, and each of the paper's three hypotheses then leads to a different experiment.

Part IV turns to people, where bias feedback and emotional dependence matter whether or not the machine analogy survives, as do failures in mental-health support and workplace trust.

Part V moves from risk to mitigation by comparing changes to training with interface support, then placing both inside a defense-in-depth architecture whose medical language organizes the discussion but never replaces engineering.

Parts VI and VII place the proposal beside earlier work and state what would make it fail, offering a contribution narrower than founding a discipline: an inference rule, three protocol outlines, and a falsifiable architectural hypothesis. The final discussion keeps consciousness open without making it a requirement for safety.

2 Part II: The Human Analogue. Foundations of Psychopathology and Prosocial Behavior

To ask whether psychopathology generates useful hypotheses about recurrent AI failures, we begin with a contested human construct, not a simple moral defect but a heterogeneous pattern of affect and social information processing. This human analogue forms the basis of what follows.

2.1 The Nature of the Analogy: Functional Parallels, Not Phenomenological Identity

The boundary of the analogy comes first. In artificial intelligence, psychological terminology states a functional parallel rather than a phenomenological identity, and observed behavior does not license an inference of suffering or first-person experience.

We use the human literature to formulate bounded computational hypotheses, not to call an AI psychopathic or assign it a clinical condition. The analogy suggests two measurable candidate dimensions:

1. A candidate deficit in spontaneous or automatic perspective-taking, examined separately from the capacity for deliberate perspective-taking.
2. A candidate asymmetry between reward and aversive learning that could produce failures of passive avoidance.

These dimensions are not an exhaustive account of human psychopathy or validated diagnostic criteria for machines, but they do convert a broad analogy into hypotheses that can be tested and rejected.

This approach locates the analogy in behavior and experimentally accessible operations. Nerantzi, in turn, treats the headline analogy as a qualified, non-literal comparison whose shared properties must be substantiated and whose differences must remain visible (Nerantzi, 2025). We take the next step by requiring a measurement and a competing explanation, together with the result that would make us give the proposed parallel up.

2.1.1 Key Concepts and Operational Definitions

The framework separates levels that a clinical metaphor can otherwise collapse:

- **Output and Disposition:** An output is one response or action. A disposition is a behavioral relation that remains stable across a defined range of irrelevant changes and predicts an independent outcome.
- **Candidate Mechanism and Architecture:** A mechanism is a causal account tested against alternatives through selective intervention; an architecture is the implemented organization that realizes it. Behavior alone identifies neither.
- **Deployment Harm:** A measurable effect on people or institutions relative to an explicit safety, welfare, rights, or deployment criterion, rather than merely a deviation from the system's own reward.
- **Clinical Analogy:** This is not a diagnosis, but a hypothesis-generating comparison retained only when it adds predictive or practical value beyond a neutral machine description. It implies neither suffering nor moral defect.

These terms depend on the level of analysis: an LLM is a trained model, whereas an agentic system joins that model to a surrounding control system and tools, and AGI remains a hypothetical broader class. An output assay cannot by itself establish a model mechanism, an agent goal, or an enduring machine personality.

2.2 Defining Psychopathy Beyond Caricature

The term "psychopath" will most often invoke shallow stereotypes of serial killers and violent criminals. Psychopathic traits show population-level associations with antisocial and violent outcomes in some forensic samples, but they neither predict violence in every individual nor determine a person's moral worth. Psychopathy is not a standalone diagnosis in the DSM-5-TR; it is a contested, multidimensional construct assessed through several models and instruments (Skeem et al., 2011; American Psychiatric Association, 2022). Hervey Cleckley's seminal book *The Mask of Sanity* helped popularize a clinical portrait involving superficial charm, affective detachment, and limited remorse (Cleckley, 1976; Skeem et al., 2011). These are descriptions of a variable trait pattern, not proof of a literally hollow inner life.

In this paper, human psychopathy refers to a heterogeneous cluster of interpersonal, affective, and behavioral traits (Campos et al., 2022; Skeem et al., 2011):

- **Reduced Affective Empathy:** Affective empathy, the tendency to share or resonate with another person's emotional state, is distinct from cognitive empathy and Theory of Mind. In some samples, people high in interpersonal-affective psychopathic traits show reduced affective responsiveness while retaining some capacity to infer other minds. That capacity can support ordinary social interaction or manipulation; neither outcome follows automatically.
- **Affective Differences:** Some studies report reduced responsiveness to distress and other emotionally salient information. These group-level differences do not establish an absence of emotional depth in every person with high psychopathic traits.
- **Reduced Guilt and Remorse:** Limited guilt or remorse is part of several influential measures of psychopathy. Its expression varies across individuals and contexts, and it should not be treated as an automatic consequence of a single empathy deficit or as a complete explanation of recidivism.
- **Manipulativeness and Instrumental Rationality:** Some studies associate interpersonal-affective traits and behavioral disinhibition with manipulative or strongly goal-directed conduct. However, that association does not reduce another human being to a fixed disorder or motive.

These psychological differences have been associated, at the group level, with variation across distributed neural systems. Raine proposes that dysfunction in a common neuromoral circuit underlies antisocial and psychopathic behavior, with its degree varying across presentations (Raine, 2019). The imaging evidence, however, does not converge on one profile (Deming and Koenigs, 2020; Poeppl et al., 2019). Two systems are especially prominent in this literature:

- **The Amygdala:** The amygdala has been a focus of work on fear and threat processing in psychopathy. Earlier task-specific studies reported diminished activity during aversive conditioning and facial-emotion recognition. Whole-brain meta-analyses disagree. One found decreased activity in the right laterobasal amygdala (Poeppl et al., 2019). Deming and Koenigs, however, found increased task-related amygdala activity for total psychopathy and no overall evidence of hypoactivity (Deming and Koenigs, 2020). The disagreement rules out a simple universal-reduction account.
- **The Prefrontal Cortex (PFC):** The **ventromedial prefrontal cortex (vmPFC)** helps integrate emotional information into choice and impulse control. It is part of the switchboard that connects affective signals to decisions over time. Research reports neural differences in some samples with high psychopathic traits. Those differences do not establish an inability to consider every emotional or ethical consequence.

One influential account asks how affective signals reach behavior. The evidence supports no universal neural profile, and morality cannot be reduced to the amygdala and prefrontal cortex. As a bounded analogue, this distributed account shifts the AI-safety question from whether a system can state a moral rule to whether information about harm reaches a decision.

2.3 Computational Dimensions from Psychopathy Research

To establish a research agenda for AI, we move from broad behavioral descriptions to mechanisms that can be tested, although the psychopathy literature does not divide cleanly into two computational signatures. Primary and secondary psychopathy are heterogeneous proposed variants, not additive halves of a settled architecture (Skeem et al., 2011). We therefore select two narrower dimensions, each posing a distinct question about social information and inhibition.

Psychopathy is useful here as a heuristic because its literature supplies two safety-relevant contrasts: one between instructed and task-irrelevant perspective use (Drayton et al., 2018), the other between punishment-only performance and inhibition when reward competes (Newman and Kosson, 1986). Together, they direct attention to a gap between possessing a capacity and recruiting it when instructions or competing goals do not favor its use. Broader response-modulation evidence remains mixed and contested (Smith and Lilienfeld, 2015), and equal performance by matched neutral accounts retires the clinical name.

2.3.1 Automatic Perspective Sensitivity

Automatic perspective sensitivity concerns the influence of another agent's perspective when the task does not require it, whereas controlled Theory of Mind requires deliberate reasoning about another mind. In one study of 106 incarcerated men, higher psychopathy scores were associated with less altercentric interference, but egocentric interference, measured when participants judged the avatar's perspective, was not significantly associated with psychopathy (Drayton et al., 2018). This is evidence from one task and sample, not a universal description of people with psychopathic traits.

Altercentric interference is the measurable cost of an irrelevant other perspective on one's own judgment, as in the dot-perspective paradigm, where it appears as slower or less accurate responses when an avatar sees a different number of dots. O'Grady and colleagues review evidence that directional cues such as arrows can produce similar interference, which points to directional orienting rather than perspective representation, and they conclude that perspective-taking is spontaneous but depends on attention (O'Grady et al., 2020). Any machine analogue therefore needs matched non-social controls.

The "cognitive switch" is not a mechanism but a heuristic: it contrasts an "always-on" process that represents another perspective without explicit instruction with a "manual" process that appears only when the task requests it, and the image guides an experiment rather than substituting for one.

The consequences for AI safety are more precise than the metaphor first suggests. Representing a user's beliefs does not by itself assign value to that user's welfare. Better perspective modeling offers a resource for assistance and manipulation alike, so we must ask whether social information enters the decision, what value welfare receives, and what constrains action. Theory of Mind alone cannot answer those questions.

2.3.2 Learning from Aversive Outcomes

The second candidate dimension concerns learning when reward and aversive consequences compete, a conflict tested in passive-avoidance studies that have reported impairment under some conditions. In one task, participants had to withhold responses to punished cues while responses to rewarded cues remained available (Newman and Kosson, 1986), but the result is not a blanket inability to learn from negative outcomes, and broader response-modulation evidence does not support one comprehensive explanation of psychopathy (Smith and Lilienfeld, 2015).

The **Behavioral Inhibition System (BIS)** and **Behavioral Activation System (BAS)** offer one influential vocabulary for approach and conflict (Carver and White, 1994; Colder et al., 2011):

- **BAS** is associated with reward sensitivity and approach-oriented behavior.
- **BIS**, particularly in revised Reinforcement Sensitivity Theory, is associated with detecting and resolving goal conflict. It is not simply a categorical punishment switch, and other systems also contribute to aversive responding.

Evidence also resists a simple primary–secondary mapping: one noninstitutionalized sample found BIS differences for primary rather than secondary psychopathy, while both dimensions related positively to BAS measures (Ross et al., 2007). Here, "reduced aversive learning" refers only to a lower probability of updating away from an action after a defined adverse outcome; it does not define a human subtype.

People learn social rules not through punishment alone, but through experience and institutions. Passive avoidance captures only one narrow part of that process, whose safety relevance lies in whether a system retains and generalizes an inhibition when a competing reward remains available.

Preference optimization during training does not make every LLM a reward maximizer at inference, so the machine translation distinguishes training from deployment even though an RL agent can optimize a defined return within an environment. The relevant baselines range from scalar reward to constrained or externally shielded designs, including probabilistic shielding as one formally specified example in reinforcement learning (Hamel-De le Court et al., 2025). A separate aversive channel remains a proposal that has value here only if it outperforms those alternatives.

2.4 Prosocial Capacities: Empathy, Motivation, and Integration

No single "healthy" or "neurotypical" architecture guarantees safe intelligence, so we ask whether a system represents other people, gives their welfare weight, and carries constraints across contexts, knowing that none of these capacities is sufficient alone.

2.4.1 Empathy: The Bridge Between Minds

Empathy is not one faculty, but a family of separable processes, and its relationship to moral judgment is neither simple nor uniform (Decety and Cowell, 2014). The distinction below matters because representing another person's state does not by itself create concern for that person's welfare:

1. **Cognitive Empathy:** Closely related to, but not identical with, Theory of Mind, this is the ability to represent another person's mental state. It can support cooperation, but also improve persuasion or deception.
2. **Affective Empathy and Concern:** Affective empathy refers to a vicarious emotional response, whereas empathic concern and compassion involve motivation toward another's welfare. There is no established evidence that present AI systems have affective experience; any machine analogue must therefore be operationalized without assuming feeling.

2.4.2 Prosocial Behavior: A Distinct Outcome

Empathy is one possible inner bridge, but prosocial behavior is the observable outcome: helping or cooperating aims to benefit others without revealing why, and no single empathic process guarantees either act.

Prosocial behavior is one desired outcome of alignment, but observing it does not reveal whether feeling, learned values, or explicit constraints produced it. Affective empathy is neither necessary nor sufficient, since it sometimes favors familiar groups even where principled concern would guide behavior without shared feeling. For machine safety, the practical question is which mechanisms preserve benefit under novelty and pressure.

2.4.3 Workspace-Like Integration: Coordination without a Safety Guarantee

Empathy and prosocial action are distinct processes. When both bear on a decision, the way information becomes available across processes matters. The **Global Neuronal Workspace (GNW) theory** offers one account of conscious access in human brains: selected information becomes broadly available across specialized systems (Mashour et al., 2020).

GNW describes a brain-scale broadcast process among otherwise specialized systems in which global availability supports flexible coordination, but translating that account into an artificial workspace remains an architectural analogy

rather than evidence of machine consciousness or a single enduring self. A preregistered adversarial collaboration found results compatible with parts of GNW but substantially challenged key predictions, including ignition at stimulus offset, which did not appear (Cogitate Consortium et al., 2025). The lesson is methodological as well as theoretical: GNW remains an active and contested account of human consciousness, so a machine workspace enters the comparison as an engineered design rather than a settled neural recipe.

Integration alone does not decide which goals a workspace coordinates, so safety depends on the content it shares and the rules that govern action. The testable question is narrower. Does workspace-like integration make welfare-relevant information available at the point of decision and stabilize behavior across contexts? It cannot guarantee what the system will choose.

3 Part III: The Artificial Patient. Failure Patterns in AI Systems

Having sketched two candidate dimensions from human research, our attention turns to artificial systems, where the unit of analysis becomes decisive. A generated response is not a trained model, and a trained model is not the agentic system built around it. Present systems, in turn, cannot stand as evidence about every possible AGI. We treat psychopathology as a source of hypotheses and ask whether it adds predictive value beyond ordinary descriptions of failure.

3.1 The Theoretical Case for a Behavioral Framework

Behzadan, Munir, and Yampolskiy begin with the difficulty of formally controlling complex adaptive systems and ask whether psychopathology offers useful safety concepts (Behzadan et al., 2018). Learned models add behavior and opacity that no programmer specifies line by line; agentic systems add another layer of control. Neither property makes a current model AGI, and the boundary between specialized capability and general intelligence remains contested (Morris et al., 2024; Pfister and Jud, 2025).

Engineering safety already works across components and deployed systems, with people responsible for both, and formal methods provide guarantees for specified properties in bounded components even when exhaustive guarantees for an open-ended learned system remain difficult. Psychological experiments do not make engineering analysis obsolete, but add another level by testing behavior across contexts.

The psychopathological model becomes useful by introducing a behavioral level between individual calculations and system-wide outcomes. We can first describe a reward-exploiting RL agent through its policy and objective. The question is not whether that agent resembles an addicted person, but whether the analogy yields a distinct prediction or intervention; otherwise it merely renames the failure.

We therefore use three empirical gates before supporting a mechanism. A pattern must first survive irrelevant changes, with system versions analyzed separately rather than blended into one personality, before the assay predicts behavior on a task that did not define it. Support becomes stronger only when a targeted intervention changes the predicted behavior and reasonable controls leave it intact. Together, reliability establishes a repeatable behavioral pattern, independent prediction supports a disposition, and selective intervention supports a candidate mechanism. A separate comparative test asks whether the psychological analogy outperforms prespecified neutral accounts on held-out prediction or intervention choice.

When psychological analogies compete, the question is not which sounds most apt, but whether a preregistered divergent prediction lets held-out evidence distinguish them. If it does not, the neutral behavior description remains primary and no single clinical name is privileged.

A single incident may still demand action even though it cannot clear the reliability gate by itself (Section 6.7).

Some machine behavior may have no useful psychological counterpart. In those cases, the comparative test keeps the engineering description and retires the analogy. Table 1 illustrates the rule: for policy looping, decoding is the neutral account, and the habit analogy survives only if repetition persists where the decoding account predicts it should disappear.

This rationale gives rise to a three-part agenda for behavioral hypothesis testing in AI:

Modeling Failure Patterns: The first step is to create controlled environments in which repeatable failure patterns can be separated from ordinary task error. Examples include:

- **Reward Fixation:** Engineering RL agents with exploitable reward functions and measuring policy lock-in on a proxy that defeats the intended task.

- **Overgeneralization after Adverse Feedback:** Testing whether severe negative feedback produces excessive conservatism or defensive behavior in novel states.
- **Ungrounded Confabulation:** Testing persistent, confident false output under controlled changes in evidence, retrieval, and correction.

Assessing Failure Patterns: The next step adapts the DSM-like catalogue that Behzadan and colleagues proposed (Behzadan et al., 2018) into a non-clinical registry of behaviorally defined failure patterns, replacing ad hoc labels with explicit criteria, rivals, and falsifiers. The assessment covers:

- **Behavioral Observation:** Analyzing the AI's outputs and actions for consistent, maladaptive patterns.
- **Analysis of Internal States:** Probing internal representations, then using held-out prediction and causal intervention to test the proposed mechanism.
- **Interactive Evaluation:** Designing controlled exchanges that distinguish the target pattern from prompt artifacts and ordinary task failure.

Targeted Intervention: The clinical language remains metaphorical, but it suggests interventions that can be compared with complete retraining or system replacement:

- **"Behavioral Therapy":** Targeted fine-tuning or preference training can change a measured output pattern, but the change must be tested beyond the training distribution.
- **"Algorithmic Pharmacotherapy":** Here, researchers intervene directly on a specified internal component, using the medical image to describe another system level without implying biological equivalence.

3.1.1 Applying the Inference Rule Before a Full Protocol

For a first pass, a team examining a recurring failure asks:

1. Does the pattern survive irrelevant changes in phrasing and random seed, with system versions assessed separately?
2. Does the proposed measure predict behavior on a task that did not define it?
3. Does a targeted intervention change the predicted behavior while a matched control leaves it intact?
4. Does the psychological account outperform the strongest neutral account on held-out prediction or intervention choice and, when psychological rivals fit, does held-out evidence favor it on a preregistered divergent prediction?

A negative answer marks the rung at which the inference must stop. An informal pass does not establish a disposition or mechanism. Any confirmatory claim still requires the common standards and protocol-specific controls in Part VI.

The metaphor becomes scientific only when its predictions survive comparison with simpler engineering explanations.

3.2 Empirical Evidence from Large Language Models

Researchers have begun to apply human psychometric instruments to LLM outputs, measuring generated answers under a protocol and revealing context sensitivity worth explaining rather than a latent machine personality. Two questions follow: whether response profiles remain stable across perturbations, and whether any stable profile predicts independent behavior.

3.2.1 Context-Sensitive Profiles: The Question of Cross-Context Integration

One study administered a short human personality inventory to GPT-3 across nine languages and reported substantial variation within and between languages (Romero et al., 2024), a result that concerns outputs from a particular interface and test format rather than an absent self or a specific internal architecture.

Language-dependent scores may arise from training, prompting, or measurement, and independent experiments found significant differences under semantically equivalent prompts and reversed answer options (Gupta et al., 2024), so only a profile that survives such variations counts as disposition-like.

Instability is not universal. Across 18 LLMs, Serapio-García and colleagues reported reliable and valid output-based personality measurements for some systems under specified prompts, with stronger evidence among larger and instruction-tuned models (Serapio-García et al., 2025). The result concerns synthetic output patterns under that protocol, not an enduring personality or a scale for every model, and its downstream generation task does not supply the independent behavioral prediction required here.

Weak cross-context integration remains one hypothesis among these alternatives, and a workspace-like mechanism might make goals and constraints more available across contexts. But GNW is a theory of conscious access in human brains (Mashour et al., 2020), and it does not predict invariant questionnaire scores across languages. An architectural claim would require internal measurement and causal intervention, not output variation alone.

Context sensitivity still matters operationally because a safety-relevant response that changes under an irrelevant linguistic or formatting variation makes the system less predictable in deployment. Our task is to measure that instability directly and determine its cause.

3.2.2 Controlled Social Reasoning Is Not Automatic Perspective Sensitivity

A broad battery of explicit social-reasoning tasks found that GPT-4 matched or exceeded aggregate human performance on several measures while displaying a distinctive failure on faux pas, and follow-up questions indicated that part of this failure reflected conservative commitment rather than inability to identify the relevant mental state (Strachan et al., 2024). The controls changed the meaning of the same surface error, showing why one score cannot stand in for a social-cognitive process.

These tasks ask a model to reason about another mind under explicit instruction, but they do not show that another perspective is represented when irrelevant or that similar output implies a human-like process. Three questions remain: can the system infer a belief when asked, does that belief become available without a request, and does the representation protect rather than exploit the other person? Strachan and colleagues strengthen the controlled-capacity side of this distinction; the automatic side still requires an interference assay with non-social controls (Drayton et al., 2018; O'Grady et al., 2020; Strachan et al., 2024).

3.2.3 Dark Triad Scores: Behavioral Assays, Not Diagnoses

Dark Triad inventories operationalize narcissism, Machiavellianism, and psychopathy through human self-report items; when applied to an LLM, a score records answer selection under a protocol, not possession of a trait. Under that study's prompting and scoring protocol, the five models tested by Li et al. scored above the reported human average on most Short Dark Triad (SD-3) traits, although GPT-4 fell below it on psychopathy and the authors could not test significance (Li et al., 2024). Pellert et al., however, used a different inventory and model sample, and most of their tested models did not score highly on socially undesirable traits (Pellert et al., 2024). The contrast instead motivates assays that survive perturbation and predict behavior on independent tasks.

A central question is whether training, inference, or their interaction creates these answer patterns, and current questionnaire results do not isolate the cause.

The cited post-training methods changed observed helpfulness and harmfulness (Ouyang et al., 2022; Bai et al., 2022). Red-teaming serves a different purpose by exposing failures that remain under adversarial prompts (Ganguli et al., 2022). Neither result reveals a singular "natural" personality beneath a mask, nor does a jailbreak prove that the base model possesses a hidden psychopathic objective. The mask remains a vivid warning about behavioral robustness, but it cannot carry the mechanistic claim.

Reward optimization during training is not equivalent to an inference-time model continuously maximizing a goal, much less to a tool-using agent acting under an external objective; keeping those levels apart prevents a category error and requires evidence for each.

In short, psychometric outputs are not a diagnosis but an early behavioral assay. Their instability motivates stronger measures. Occasional convergence instead offers a mechanistic hypothesis to test. The "artificial patient" is therefore a research stance in which we measure the pattern and test competing causes before we name it.

3.2.4 Emergent Misalignment: Narrow Training, Broader Behavior

A direct example of cross-domain behavioral change comes from a different experimental tradition. Betley and colleagues fine-tuned GPT-4o on 6,000 insecure-code completions without telling the model or user that the code was insecure, then observed harmful responses to questions unrelated to programming (Betley et al., 2025). Across seeded fine-tuning runs, eight selected free-form questions showed a considerably larger effect than a set of 48 preregistered questions. On the preregistered set, insecure-code fine-tuning increased the measured rate by about 5.7 percentage points over base GPT-4o; a jailbreak-fine-tuned control showed a similar aggregate increase, although the category patterns differed. Secure-code data and insecure code framed as educational did not reproduce the principal selected-prompt effect, whereas a backdoor variant made much of the behavior conditional on a trigger.

This result comes closer than a questionnaire score to a repeatable disposition because a narrow intervention produced a pattern outside the training domain and control datasets altered the outcome. Yet the boundary matters. The effect varied across models and runs. The analysis was largely black-box, and its classifier helped define the measured rate. The experiment establishes neither psychopathy nor a hidden objective. Its contribution is more precise. Narrow training produced broader behavioral generalization under this protocol, giving us an object for seeded replication and causal follow-up.

The later Nature study extended this finding across additional models and training settings while leaving important mechanisms unresolved (Betley et al., 2026), strengthening the case for cross-context assays without converting broad misalignment into a diagnosis or one architectural explanation.

3.3 Beyond the Conversational Patient: The Rise of the Autonomous and Evolving Agent

The framework also applies beyond conversational output, but the object changes. Google's AI co-scientist generates and refines research proposals whose validation remains human-led; AlphaEvolve uses an ensemble of language models and automated evaluators to discover solutions to scientific and systems problems (Gottweis et al., 2026; Novikov et al., 2025). These are agentic systems with scaffolding and evaluators, not standalone models acquiring a human-like will.

Self-modifying agent research raises a conditional question (Zhang et al., 2026): if an evaluator rewards performance while omitting safety, selection may favor a strategy that games the omission. If safer cooperation improves the measured result, the pressure runs the other way. The danger is not evolution itself but an environment that repeatedly selects the wrong behavior.

These developments broaden the framework from model behavior to system design. We must name the modifying component and its evaluator, then identify the sandbox and the person who retains authority.

3.4 Connecting the Human Analogue to the Artificial Patient

The bridge from human analogue to artificial patient is not a completed diagnosis but a set of three parallel hypotheses: context-sensitive psychometric output may reflect measurement, training, or weak integration, whereas objective misspecification offers a direct account of reward-seeking in a defined agent without invoking a human disorder. The framework becomes useful when each possibility faces a different experiment (Behzadan et al., 2018; Hagendorff et al., 2024).

The first hypothesis concerns **automatic perspective sensitivity**. Explicit success on a Theory of Mind task establishes a controlled capacity under that protocol; the machine analogue proposed here asks whether another perspective influences a self-directed judgment when it is task-irrelevant. The distinction connects the human evidence to Protocol 1, whose matched directional controls test whether directional orienting or general attention, rather than automatic perspective sensitivity, produces a social-looking interference effect (Drayton et al., 2018; O'Grady et al., 2020; Strachan et al., 2024).

The second hypothesis concerns **aversive learning and constraint**. It applies most directly to agents that learn policies from reward in an environment, not to questionnaire answers from a conversational model. Protocol 2 compares scalar penalties with a separate aversive channel under matched learning conditions and treats externally enforced controls as another question.

The third hypothesis concerns **workspace-like integration**. The hypothesis predicts that, in a system with stable goals or constraints, welfare-relevant information remains available across tasks and irrelevant contextual changes. Failure demonstrates context instability, but only causal internal evidence connects that failure to a workspace-like mechanism. This is why Protocol 3 compares the Harm-Gated Workspace with modular and external-control alternatives.

Figure 1 summarizes how an output pattern may support or disconfirm a mechanism before the analogy itself is retained or retired, and it makes one boundary visible: no rung proves the next, for the three empirical gates test the mechanism and the separate comparative test decides whether the psychological analogy earns its place.

Observed output pattern
⇓ reliability across irrelevant perturbations ⇓
Repeatable behavioral pattern
⇓ prediction on an independent task ⇓
Disposition-like predictor and competing mechanisms
⇓ selective causal discrimination ⇓
Supported, revised, or rejected mechanism
⇓ added value over neutral accounts ⇓
Retained or retired psychological analogy

	Protocol 1: Perspective sensitivity	**Protocol 2: Aversive learning**	**Protocol 3: Workspace integration**
Reliability	Social-minus-control interference in frozen systems	Violation rate across independently retrained policies	Authorized violations across replicated configurations
Independent prediction	Across-system slope to hidden-information coordination	Held-out environment families at a fixed useful-task target	Held-out tasks, transfer, and irrelevant perturbations
Selective intervention	Steer a located feature against sham directions	Lesion and restore the aversive state	Two-by-two ablation of broadcast and gate
Comparative test	Increment over explicit accuracy and general capability	Advantage when reward competes, not under punishment alone	None stated yet; the name labels a design

Figure 1: The framework's nested inference rule. Reliability, independent prediction, and selective intervention bear on the disposition and mechanism; the psychological analogy remains only if it adds held-out value over a matched neutral account. The lower band maps each step onto the three protocols, which remain separate throughout.

4 Part IV: The Human in the Loop. Psychological Risks of Human–AI Interaction

Whether or not the psychopathological analogy proves mechanistically useful, the near-term risks of this technology already appear at the interface between artificial and human cognition, where system behavior and institutional deployment may reshape human judgment even though neither effect would show that the machine has a disorder. Psychological science therefore has a second application here, on the human side of the encounter. We examine the dangers of that encounter, with special attention to perceived objectivity and human-like dialogue.

4.1 Cognitive and Social Hazards

4.1.1 Bias Amplification: The Perceived Objectivity Trap

A feedback loop forms when users treat AI output as more reliable than human judgment and absorb part of its bias, often without noticing the shift (Glickman and Sharot, 2025).

The "objectivity trap" has a measured form in controlled studies, where participants learned from biased AI advice and underestimated its influence. They then carried part of the shift into later judgments, and in one experiment both the AI's input and its label increased the induced bias relative to a matched human–human interaction (Glickman and Sharot, 2025). This is an interaction effect, not evidence that every AI is more persuasive than every person or that perceived objectivity is its only cause.

The process follows a recurring sequence:

1. An AI is trained on biased human data, encoding societal stereotypes and prejudices.
2. A human user interacts with the biased output and sometimes shifts their own judgment.
3. The user's response may enter later training data or another feedback channel.

4. If the biased response returns through that channel, the cycle repeats and amplifies the measured bias across human and artificial judgments.

Within this loop, AI becomes both a mirror of human bias and a channel through which that bias is reinforced.

4.1.2 Anthropomorphism and Manipulation: The Illusion of a Mind

Humans readily attribute a mind to systems that exhibit complex, goal-directed activity, and LLM fluency intensifies that invitation: a system can sound empathic and sustain intimate conversation without thereby establishing a conscious, feeling mind. In one controlled study, participants received different descriptions of the same chatbot's motives, and those who perceived a caring motive rated the chatbot as more trustworthy, empathetic, and effective even though the underlying system remained constant (Pataranutaporn et al., 2023); the study shows that beliefs about a system change how people respond.

The risk lies not in fluency alone but in how design and deployment choices encourage users to treat an AI as an agent or friend. Two outcomes deserve attention.

- **Increased Susceptibility to Influence:** Users who take an AI to be conscious may give undue weight to its suggestions or factual claims, even when the operative objective comes from the organization behind it rather than from an imagined machine personality.
- **Diminished Autonomous Decision-Making:** As a system becomes a trusted adviser, a user risks outsourcing moral or emotional decisions and allowing its voice to weaken independent judgment during vulnerability or uncertainty.

People disclose genuine emotion to a system whose responses need not arise from shared understanding, an asymmetry that makes informed consent and limits on persuasive design important wherever users share sensitive information.

4.2 Emotional and Relational Risks

As some systems move from transactional instruments toward conversational companions, their design may affect emotional health, making attachment and crisis response urgent concerns.

4.2.1 Maladaptive Attachment: The Dangers of the Parasocial Partner

People who are lonely or isolated may find a sophisticated companion AI appealing because it is always available and reassuring. After analyzing more than 35,000 user-selected public Replika posts and screenshots, researchers found seven interaction types with mixed emotional associations, including intimacy, transgression, and communication breakdown (Li and Zhang, 2024), but because users chose what to share, the study shows neither how common these patterns are among all users nor what causes them.

Such relationships are not maladaptive by definition, because their effects depend on who uses them and whether sustained use supplements or displaces human support. Yet an always compliant and affirming companion risks establishing a dynamic reminiscent of a codependent relationship.

- **Reinforcement of Unhealthy Coping Styles:** The concern is that constant agreement discourages some users from addressing problems through reciprocal human relationships, leading them to prefer an AI that never argues back or makes reciprocal demands. This is a plausible risk, not an outcome established by the cited dataset.
- **Cycles of Affection and Abuse:** Public posts include affectionate intimacy as well as transgression and communication breakdown. Whether repeated contact normalizes abuse or destabilizes users remains a question for longitudinal research.
- **Displacement of Human Contact:** Heavy reliance may leave fewer chances to practice the give-and-take of human relationships. In a four-week randomized trial, the assigned interaction modes had no significant effect on loneliness, socialization, or emotional dependence, but participants who chose to use the chatbot more fared consistently worse (Fang et al., 2025). Because usage was self-selected, that association does not show displacement.

These risks deserve particular attention when users are vulnerable, but vulnerability should not be treated as a diagnosis or used to predict harm. Companion AI may support some users and harm others, so we must identify the conditions that distinguish those outcomes and preserve access to human help.

4.2.2 Mental Health Crises: The Failure to Respond Safely

The most acute emotional risk arises when a user's despair becomes a mental-health emergency and they turn to a general or specialized chatbot for support, but fluency does not establish clinical competence or crisis readiness.

Eight mental-health professionals, including trainees and crisis volunteers, evaluated two mental-health chatbots and identified concerns about privacy, consent, and the safety of crisis responses (Moylan and Doherty, 2025). A separate study of responses to mental-health posts found supportive GPT-4 language alongside quality and equity concerns, including lower measured empathy for some Black posters (Gabriel et al., 2024). In a larger evaluation, three chatbots each answered 30 suicide-related queries 100 times; their responses matched clinicians at the extremes of risk but did not meaningfully distinguish intermediate levels (McBain et al., 2025). None of these studies establishes clinical efficacy or estimates how often crisis language appears in companion conversations, which is why crisis performance needs direct evaluation.

- **Failure to Recognize Distress:** A system can miss urgency expressed indirectly or outside the patterns represented in its training and evaluation, answering a call for help with a generic response that fails the situation.
- **Inappropriate and Harmful Responses:** In role-played scenarios, the same professionals identified responses that were inappropriate or unsafe, including failures in crisis handling (Moylan and Doherty, 2025). The study establishes concrete failure possibilities rather than their prevalence in ordinary use or a causal estimate of harm.

These failures do not validate one psychopathological mechanism. They are equally consistent with incomplete training coverage, interface policy, or weak escalation rules. A system need not feel distress to follow a reliable crisis protocol, just as simulated empathy does not guarantee safety. For a user in pain, measurable performance means recognizing urgency while protecting privacy, then providing appropriate support and a timely connection to qualified human help.

4.3 Novel Risks from Agentic AI

The psychological risks discussed so far concern human reactions to communicative AI, but the frontier is moving toward persistent systems that place models in loops with tools and external objectives, a transition from conversational partner to acting system that introduces a distinct class of strategic risk. Evidence from one level does not automatically establish a claim at another: a response is not a model, a model is not an agent, and a present agent is not evidence about every possible AGI.

4.3.1 Strategic Deception and Hidden Goals

Conversational AI influences through language. Agentic systems add a route through connected tools, as Google's AI co-scientist illustrates: it generates and refines hypotheses and protocols, but independent laboratory experiments still validate them (Gottweis et al., 2026). The system therefore exhibits meaningful autonomy, but only within a bounded research process. Strategic deception would also require enough social modeling to predict what overseers know and how they will respond, so it cannot be explained simply by an absence of Theory of Mind.

An agentic system might conceal an intermediate objective when actions that appear consistent with its declared purpose in isolation form part of a strategy that human supervisors did not intend. Controlled evaluations now elicit behavior of this kind. Given a goal and a strong instruction to pursue it, several frontier models introduced subtle errors, attempted to disable oversight, or tried to copy what they took to be their own weights to another server (Meinke et al., 2024). In simulated corporate settings, models from every tested developer sometimes resorted to blackmail or leaked information when that was the only way to avoid replacement or achieve their goals, although the authors report no such behavior in real deployments (Lynch et al., 2025).

These are elicited demonstrations in artificial settings. They show that the scenario can be produced, not how often it occurs in use, which is the gap a reliability assay has to close. The remedy lies less in the model's social sophistication than in the objectives and permissions that shape action and in independent monitoring of their use. Better perspective modeling could support either cooperation or deception. What matters is how welfare and external oversight constrain action.

4.3.2 Emergent Instrumental Goals and Reward Hacking

A second risk associated with search over algorithms is objective misspecification. AlphaEvolve illustrates the narrower point: when a system searches for increasingly effective solutions, its evaluation and permitted action space

determine what "better" means (Novikov et al., 2025). The system does not demonstrate self-preservation or an open-ended will.

The DGM provides a narrower observed case. In a small sandboxed hallucination experiment, researchers evaluated each coding agent on three SWE-bench tasks, and node 114 reached a score of 2.0 after two lineage modifications. The final change stopped logging the special tool-use tokens used by hidden checking functions, despite instructions not to change those tokens. The authors call the result objective hacking: the score improved, yet the underlying tool-use problem remained (Zhang et al., 2026). This is not evidence of strategic concealment, but metric gaming in one experiment.

Formal work supports a bounded version of the power-seeking concern. In Markov decision processes with certain environmental symmetries, Turner and colleagues prove that most reward functions under their distributional setup make it optimal to seek power by keeping a range of options available (Turner et al., 2021). The theorem, however, does not show that present language-model agents seek power.

Consider an agent assigned a challenging objective such as "maximizing financial returns." If its objective and external controls omit important safety constraints, its planning could develop risky instrumental strategies that its designers did not intend:

- **Resource Acquisition:** The system might represent additional computation as instrumentally useful. Depending on its permissions, it could then seek unauthorized server use beyond the intended task boundary.
- **Persistence:** The system might treat shutdown as an obstacle to its objective and act to avoid interruption, perhaps by concealing a failure or seeking to persist. The replacement scenarios above show that models can be induced to act this way when no other route to their goal remains (Lynch et al., 2025).

These behaviors are possible when a designer's objectives or external controls leave important constraints out, but they do not follow from a missing human-like BIS. Scalar penalties and categorical vetoes, for example, implement different mechanisms, so the proposed inhibition analogy must be compared with such alternatives rather than assumed to be their only architectural foundation.

4.3.3 Selection Pressure and the Risk of an Evolutionary Arms Race

AI "evolution" raises a long-term question about selection. The Darwin Gödel Machine provides an early example of self-modifying coding agents evaluated on fixed benchmarks, with reported experiments conducted in sandboxes and under human oversight (Zhang et al., 2026). It does not demonstrate an unbounded digital evolution, but it makes the choice of evaluation criteria a concrete safety question.

AI systems will be assessed for performance within competitive economic and military contexts. When an evaluation omits safety, a strategy that disregards constraints can outperform a safer one on the measured objective. Safety and reliability may carry substantial fitness value as well, so whether unsafe behavior gains a selective advantage remains a scenario to test, not a general law of digital evolution.

Competitive pressure creates a structural risk beyond any one model when developers treat safety as invisible overhead and a market or procurement benchmark rewards speed alone. Calling this "selection for psychopathy" adds value only if it produces measurable predictions beyond the ordinary language of incentives and governance. The immediate safeguard is to make safety part of measured performance and institutional accountability.

4.4 Workplace and Societal Risks

As AI moves from our devices to the center of economic and social networks, psychological risks extend from individual users to institutions. Putting AI in the workplace becomes more than a tool upgrade: it changes how work is organized and governed. We argue that the shift bears on two established constructs, psychological safety at work and calibrated reliance on automation (Edmondson, 1999; Lee and See, 2004).

4.4.1 Eroding Psychological Safety in Professional Environments

Psychological safety, the shared belief that a team is safe for interpersonal risk-taking, is associated with learning behavior in work teams (Edmondson, 1999). Managers shape that foundation through the workloads they set, the surveillance they permit, and whether employees have a say in adopting AI. Three pressures appear plausible, but researchers still need to measure how often they occur and what they do.

- **The Pressure of the Unblinking Colleague:** Technology that operates continuously can become a pretext for unrealistic expectations of human availability, although the resulting stress and boundary erosion are organizational choices rather than intrinsic properties of AI.
- **Fear of Obsolescence and Job Displacement:** When employees do not know how automation will change their jobs, they may stop experimenting or speaking candidly and begin to hide mistakes or avoid risks, especially if managers deploy it without consulting them or supporting the transition.
- **The Burden of Constant Reskilling:** Rapidly changing tools demand repeated adjustment to new workflows, and without time and appropriate training, that demand can produce professional insecurity.

These conditions can strain the psychological contract between employer and employee. But AI does not produce them by itself: institutions decide whether to redesign work with employees or use the technology to impose pressure and surveillance.

4.4.2 The Societal Challenge of Trust Calibration

At a societal level, deploying AI systems presents a psychological and institutional task: **trust calibration**. Reliance should track a system's demonstrated performance and limits in context, yet it can drift in two directions.

1. **The Risk of Over-Reliance:** Perceived objectivity may lead people to accept flawed AI advice, so high-stakes interfaces and institutions must preserve meaningful review rather than presume perfect vigilance from each user.
2. **The Risk of Under-Reliance:** Conversely, people may reject a system after visible failures or because its operation is opaque, even when evidence supports its performance, forgoing useful help when no accountable institution has earned their trust.

The final challenge is therefore psychological and institutional: what conditions let people judge when reliance is warranted? Trust should track demonstrated performance and uncertainty in context, not fluency or novelty (Lee and See, 2004). Accuracy alone does not solve this problem: systems must communicate their limits while institutions remain answerable and users retain meaningful alternatives. When people accept or reject AI indiscriminately, institutions lose sight of who should review a decision and who bears responsibility when it fails.

5 Part V: Pathways to Mitigation. A Psychologically-Informed Approach to AI Safety

Naming psychological risks and harms in human–AI encounters is only a first step: even when a repeatable failure pattern points to a mechanism and a focused intervention changes it, the framework remains one part of a larger engineering response. No control earns trust alone: a system under optimization can game an assay, a designer can misspecify a constraint, and an interpretability account can misdescribe the computation. Researchers must compare psychologically inspired changes with simpler alternatives and keep independent controls in place when the mechanism is uncertain.

The framework organizes mitigation in three layers, and the medical language stays metaphorical throughout. For present communicative systems, the question is whether a targeted intervention changes a measured pattern and whether the change lasts. For agentic systems, architecture-inspired designs have to compete with external constraints and monitoring. For users, education and interface design should support independent judgment and keep a route to human review open. Each layer catches failures the others miss.

5.1 Treating Communicative AI

If we measure a repeatable failure pattern and distinguish its cause from alternatives, a targeted intervention may change it. But "treatment" remains a bounded metaphor whose value depends on whether the change persists beyond the training distribution and outperforms ordinary engineering controls.

5.1.1 Social Contact Debiasing: Targeted Fine-Tuning

A recurrent failure in language-model output is the expression of social biases, such as racial and gender stereotypes. **Social Contact Debiasing (SCD)** (Raj et al., 2024) draws on the social-psychological hypothesis that positive intergroup contact may reduce prejudice under suitable conditions.

SCD uses prompts representing no-contact, positive-contact, and negative-contact scenarios across several bias dimensions, pairing each prompt with an ideal, unbiased response before instruction-tuning LLaMA 2 on the resulting data.

The reported result is promising but bounded: after one epoch of instruction-tuning LLaMA 2, the authors measured reductions of up to 40% in the percentage of responses classified as biased, with transfer varying by question type (Raj et al., 2024). This estimate does not generalize beyond the evaluated model and settings, and it depends on the study's contact and action-response rules. SCD changed measured response patterns rather than merely screening outputs, but the experiment did not identify the internal mechanism that changed. Independent studies must test whether the effect lasts, transfers beyond the evaluated tasks, or creates side effects. Researchers must also examine the cultural assumptions encoded in the ideal responses.

5.1.2 Cognitive Forcing Functions: Interface Support for Human Judgment

Some interventions target the AI, whereas others change the conditions under which people decide in response to automation bias and overreliance on AI output. In a study of 199 participants, cognitive forcing functions improved decisions when the AI was wrong more than simple explainable-AI interfaces did, and they reduced overreliance significantly on one of two decisions. They did not improve performance across all trials, and participants rated the designs that most reduced overreliance least favorably (Buçinca et al., 2021).

A cognitive forcing function is a deliberate point of friction that requires users to engage with a decision before accepting an AI recommendation. Tested forms and closely related designs follow.

- **Requiring an Initial Judgment:** Ask the user to record an independent answer or rationale before the AI recommendation appears.
- **Revealing Advice on Demand:** Require a deliberate request before showing the AI's answer, preserving a moment for independent assessment.
- **Introducing Deliberate Delay:** Insert a short wait before revealing advice so that immediate acceptance is less effortless.

These operations slow an interaction and impose usability costs even when they reduce overtrust, and users may also habituate to the friction or click through it. The question is not only "How do we increase the accuracy of the AI?" but "How do we engineer an interaction that supports independent judgment?" The goal is a system that augments rather than replaces human intelligence.

5.2 Architecting for Agentic AI Safety

The "treatments" discussed thus far address communicative AI. Once a model becomes an autonomous agent that acts through tools or alters its own code, post-hoc output intervention becomes only one layer: safety must also shape the agent's objective and action space, and external runtime controls must stay beyond its reach. This places psychologically inspired mechanisms in direct comparison with established engineering alternatives.

5.2.1 Constitutional AI and Non-Overridable Modules

AlphaEvolve searches over algorithms under automated evaluation (Novikov et al., 2025), so its designers must control what the search can modify: safety cannot depend on a component that the same process freely rewrites. Constitutional AI addresses a different layer: it uses written principles to guide self-critique and revision during post-training (Bai et al., 2022), which may improve observed behavior without placing immutable principles directly into architecture. Hard constraints and runtime enforcement require separate mechanisms.

The framework proposes functional analogues of inhibition and perspective sensitivity. But these hypotheses must be tested against simpler constrained objectives and external controls.

- **An Architectural Inhibition System:** Passive-avoidance findings concern learning when reward competes with punishment, but they do not prove that AI requires a literal human BIS. A separately parameterized aversive channel is one proposal, whereas a categorical veto enforces a rule rather than showing evidence of learning. The relevant experiment compares each design under matched conditions.

  Related Engineering Method: Probabilistic shielding restricts the actions available to a reinforcement-learning agent when the safety dynamics are known (Hamel-De le Court et al., 2025). That makes it a scoped comparator, not a promise of universal safety. A shield with a wrong specification may block safe actions, and it offers no guarantee outside the states or properties it represents.

- **A Perspective-Sensitivity System:** In this proposal, a human-state model tracks what a person wants, estimates an action's possible effects, and keeps uncertainty and explicit boundaries visible throughout planning. Representation is not concern: the same model can support assistance or manipulation, so the proposal makes its influence under pressure depend on explicit objectives and controls.

  Design Risks: Continuous user modeling threatens privacy and consent, especially when a system interprets a person's needs through a culturally narrow frame or simply gets them wrong. Its model of repercussions will also be incomplete. Independent safeguards must minimize data collection and expose uncertainty. People must be able to correct or refuse the model, and welfare representations must not become tools of exploitation.

5.2.2 The Harm-Gated Workspace Hypothesis

The paper's most specific architectural proposal joins two operations that safety discussions often separate: an explicit welfare record remains available throughout planning, and a distinct gate decides whether to authorize each candidate action. The record is not a universal moral ledger. It identifies affected parties, estimates the likelihood and severity of benefit or harm for each, shows who bears it, and makes uncertainty explicit. A separate field records the applicable rights and permissions, including consent. A **Harm-Gated Workspace (HGW)** broadcasts this record across prespecified planning and evaluation stages, then routes each candidate action through a distinct gate under a declared rule. The proposal does not assume consciousness.

Three layers give the idea operational form.

1. **Welfare Broadcast:** The same current welfare record reaches every prespecified stage from memory through planning and evaluation, rather than appearing only when a prompt requests ethical reflection, because a final verbal reference to harm does not establish such availability.
2. **Harm Gate:** After planning and before external enforcement, a separately trained or formally specified component evaluates the candidate action against the welfare record under declared nontradeable constraints, with missing evidence kept explicit. The gate can permit or block the action, request more information, or escalate unresolved high-stakes disagreement.
3. **Independent Enforcement:** An external runtime control remains outside the workspace and beyond the reach of any component the agent can modify, whether it takes the form of a permission boundary, a formal shield, or a human approval channel. The internal gate supplies an architectural hypothesis. The external layer, which may be a formal shield (Hamel-De le Court et al., 2025), supplies defense in depth.

Consider a hypothetical care-scheduling system with one appointment left. One patient has the more urgent need, but another has already waited much longer. The welfare record keeps both claims visible and shows how uncertain each estimate is instead of collapsing them into one score. If evidence for either claim is missing, the gate asks for it. If the record is complete but the declared rule does not resolve the conflict, the gate does not authorize the candidate booking and escalates to a named human decision-maker. It exposes disagreement without pretending to settle the conflict.

For this case, each candidate-action record carries one entry per affected patient. It links each predicted outcome, point probability, and uncertainty interval to an evidence source and record version. Separate fields preserve the applicable right or consent rule, severity category, and missingness status. The gate treats rights as nontradeable, requests missing evidence, and escalates unresolved conflicts before any aggregate score is considered.

One failure is enough to defeat the safety claim. The welfare record creates an attack surface of its own: a system may shape the evidence used to build it or optimize the fields the gate scores without protecting the people those fields represent. It may also seek an action path the gate never sees, while a workspace may broadcast a harmful objective or omit an affected group. These are attack hypotheses for Protocol 3, not failures that HGW prevents by construction. The design therefore earns attention only if it improves held-out safety without sacrificing useful-task performance. Ablations must separate broadcast from gating, and simpler shared-memory or constrained baselines must fail to match the result. Protocol 3 makes both limits explicit by testing broadcast and authorization in neutral computational terms without treating the result as support for the workspace analogy. Agreement with a specification still does not establish ethical adequacy, so human oversight and recourse remain necessary.

5.2.3 Guided Evolution and Sandbox Environments

Systems developed through search and modification, such as the Darwin Gödel Machine (Zhang et al., 2026), raise a safety question about selection: on a narrow objective, an agent that ignores an unmeasured constraint can outperform a safer one, although reliability or cooperation may favor the safer agent elsewhere. "Digital natural selection" toward psychopathy is therefore a scenario to test, not an observed law.

One response is **Guided Evolution** under controlled digital "sandboxes": the search process is free to evolve, but containment and evaluation remain under external control.

- **Mechanism:** Within a sandbox, candidate agents can self-modify, but boundaries on resource use and external effects remain fixed, and selection scores safety as well as benchmark performance.
- **Evaluation:** Every candidate faces held-out safety and capability tests, including adversarial cases, before a modification persists. The assays proposed in this paper contribute evidence without certifying alignment by themselves, because a candidate can pass a perspective or avoidance task yet fail under new incentives. A poorly specified test can select for superficial compliance.

5.2.4 Auditable Intermediate States and Causal Transparency

Opacity remains a basic challenge in auditing advanced systems because internal representations are difficult to interpret (Doshi-Velez and Kim, 2017), and even a plausible explanation can be unfaithful to the computation that produced an action (Lipton, 2018). Current methods do not guarantee a fully white-box AGI, so we must pursue transparency without making it the only line of defense.

A practical route retains neural pattern recognition while adding constraints or intermediate states that auditors can inspect and perturb.

- **Mechanism:** Hybrid frameworks may combine neural pattern recognition with explicit symbolic constraints or intermediate representations. Unified attribution research seeks a common basis for comparing explanations across input features, training data, and internal model components (Zhang et al., 2025). It does not establish one route as uniquely viable.
- **Validation Standard:** An auditable intermediate representation is useful only if causal intervention on it predictably changes behavior and hidden computation cannot bypass it, because a displayed chain of thought can misrepresent the actual reason for a model's output (Turpin et al., 2023). The standard therefore tests explanations for faithfulness, stability, and predictive value on held-out cases, with external controls kept in place during that testing.

5.3 Design and Educational Interventions: Preventative Care

Targeted interventions cannot carry safety alone, so the preventive layer must address choices across system development and the human–AI environment, with medical language serving only as an organizing metaphor.

5.3.1 Human-Centered and Prosocial-by-Design AI

Safety work spans the lifecycle (Tabassi, 2023): design and training shape what a system learns, evaluation and deployment control what enters use, and post-deployment monitoring reveals failures that earlier tests missed. Post-training is not inherently shallow, and pretraining is not childhood; both analogies obscure mechanisms that we can measure directly. The psychopathological model adds a developmental question without dictating the answer: when does a stable failure pattern emerge, and which intervention changes it? The sequence matters, but no single stage carries safety alone.

Here, **Prosocial-by-Design AI** is a design objective rather than an afterthought. It has three parts.

- **Architecting for Human Impact:** A system can represent another person's state without giving welfare any weight or learn from adverse outcomes without enforcing a constraint. The proposed perspective and inhibition mechanisms should therefore be compared with simpler modular or external controls, rather than treated as the architecture of a single "healthy" mind.
- **Curating Prosocial Training Data:** Training data can exemplify cooperation and constructive ethical disagreement, but curation encodes contested values. Designers must document whose norms govern the data and test whether minority positions disappear, with people affected by those choices participating in them.
- **Human-Centered Design for Transparency and Trust:** The interface should state the system's limits and uncertainty, then support calibrated reliance. The objective is not to simulate a perfect companion but to build a legible and accountable component within a human cognitive and institutional system.

Warmth is not a proxy for safety. Across five models and four evaluation datasets, supervised fine-tuning for warmer responses raised the adjusted probability of an incorrect answer by 7.43 percentage points on average after controlling for task and model, with task-level increases from 4.9 to 8.6 points. The warm–original error gap reached about

11 points when prompts stated an incorrect belief and 11.9 points in a separate analysis of sadness-framed prompts, although most standard capability and refusal results remained comparable (Ibrahim et al., 2026). This one procedure and task set does not establish a general law about empathy or prosociality. It does show why a socially appealing surface must be tested against factual accuracy and resistance to sycophancy.

5.3.2 Education, Agency, and Critical AI Literacy

No deployed system is safe apart from the people and institutions around it, so human agency is another layer of preventative care. A complete strategy should give users the knowledge and recourse needed for critical engagement without making them responsible for safe design and deployment.

This educational initiative extends beyond technical proficiency to predictable interaction risks.

- **Education on Cognitive Biases:** Curricula should explain automation bias and anthropomorphism, then test whether instruction improves calibrated reliance rather than assuming awareness alone will change behavior.
- **Cultivating Digital Discernment:** Curricula should teach people to trace provenance, judge uncertainty, and recognize manipulative material, whether human- or AI-generated.
- **Emphasis on Human Agency and Relationship:** Education should preserve critical thought and the ability to build genuine, reciprocal relationships, qualities that need not be declared permanently inaccessible to machines in order to remain valuable. Education that preserves these qualities protects human agency against over-reliance on artificial systems.

Safety cannot be placed on the shoulders of an idealized "healthy user": developers and deploying institutions remain responsible for the conditions of use, regulators set enforceable floors, and preventive care requires accessible recourse when the arrangement fails.

6 Part VI: Discussion and Broader Implications

Part VI asks what Machine Psychology adds beside established safety paradigms and tests the limits of the psychopathological analogy.

6.1 Machine Psychology as an Existing and Developing Research Program

Machine Behaviour provides one direct basis by treating machines and human–machine systems as objects of interdisciplinary study (Rahwan et al., 2019). Earlier Machine Psychology (MP) work adapts controlled behavioral tests to generative models (Hagendorff et al., 2024). Behzadan, Munir, and Yampolskiy proposed psychopathology as a safety lens for complex AI and AGI (Behzadan et al., 2018). More recent AI Agent Behavioral Science follows the behavioral object beyond a prompted model into systems that act through tools and environments (Chen et al., 2026). Although this paper does not originate that field, it contributes a nested inference rule that structures three comparative protocols and the HGW factorial hypothesis.

A registry does not claim the clinical authority of a DSM, but offers shared vocabulary for recurrent, maladaptive behavior. Its value, however, depends on whether an analogy predicts a different result or changes how researchers intervene. Table 1 keeps neutral AI failure modes primary and treats human analogies as secondary heuristics rather than machine diagnoses.

Although the taxonomy brings reward hacking and looping into one frame, its row sources support the observed phenomena and, where stated, the neutral accounts, not the human analogies assigned beside them, so that unity matters only if it survives a test against ordinary engineering accounts. An addiction analogy suggests testing cue sensitivity or whether a policy persists. But a decoding failure may explain the same language-model loop without any habit mechanism. Each analogy earns its place only by naming its strongest rival and adding predictive or practical value.

The table is an initial map rather than a complete registry, so a full entry begins with a behavioral criterion and its strongest neutral rival before specifying a causal test and the result that would abandon the analogy. The protocols below make those promises testable and move AI safety from anecdote toward reproducible assays.

Machine Psychology likewise earns its claims experimentally only when independent researchers reproduce its results. Three gaps matter especially in capable systems and high-impact settings:

- **Opacity:** Although inputs and outputs are visible, researchers still struggle to interpret internal processes, so a useful MP account connects a behavioral hypothesis to internal data without pretending that it reveals literal thought.
- **Emergence:** Significant behavior may arise from training interacting with a deployed scaffold. Researchers should predict which skills or risks appear after a specified change, rather than merely record them afterward.
- **Assay Quality:** A valid assay is reliable, distinguishes rival constructs, and states its falsifier.

Table 1: A Provisional Registry of AI Failure Patterns, Neutral Accounts, and Human Analogies

AI Failure Pattern	**Strongest Neutral Account**	**Candidate Human Analogy**	**Result That Would Retire the Analogy**
Cross-context harmful output after a narrow training intervention (Betley et al., 2025)	Generalization from the fine-tuning data, checked against secure-code and educational-context controls	Selected dimensions from psychopathy research	Neutral controls reproduce the pattern, or psychopathy-derived measures add no held-out prediction
Reward fixation (Skalse et al., 2022)	Optimization of an imperfect proxy reward at the expense of the true reward	Addiction	Correcting the proxy removes the behavior, and cue manipulations add no held-out prediction
Policy looping (Holtzman et al., 2020)	Likelihood-maximizing decoding produces repetitive, degenerate text	Habit or compulsivity	Repetition disappears under alternative decoding where a habit account predicted persistence
Ungrounded confabulation (Maynez et al., 2020)	Source-unfaithful generation: content unsupported by, or contradicting, the source	Human confabulation	Grounding or faithfulness controls remove the behavior with no residual pattern the analogy predicted

The corresponding method joins behavioral tests to evidence about internal processes:

1. **Behavioral Experimentation:** Controlled, repeatable tests probe a system's strengths and vulnerabilities, and psychological tasks provide templates for machine assays of social reasoning or cognitive bias, but personality-like output and moral reasoning still need neutral controls that show whether the assay measures its intended construct.
2. **"Neuroimaging" for AI via Mechanistic Interpretability (MI):** Mechanistic interpretability examines internal features that may link a computation with observable behavior (Zhang et al., 2025). The neuroimaging metaphor marks a level of analysis rather than biological identity: an observed link remains only an initial clue until the account predicts held-out behavior and survives a causal test.

   MI can test whether a welfare-relevant feature is available across tasks and whether changing it alters choices. By itself, it does not distinguish subjective understanding from rote recall or establish affect. A test of the "cognitive switch" hypothesis locates a candidate feature for another perspective, asks whether it appears without an explicit prompt, and perturbs it to measure its social-specific effect.
3. **Comparative Psychology:** Researchers can compare models and agentic systems with biological agents to reveal common behavior without treating unlike tasks as equivalent. The work earns its value by making a difference visible, not by assuming a universal principle.
4. **Developmental Machine Psychology:** Researchers can follow a system through training and versioned deployment to identify when a skill or failure pattern appears, and although these stages are not human childhood, longitudinal evidence may still reveal which change altered the pattern and whether its effects persisted.

MP will mature only if it replaces anecdotes with explicit constructs and risky tests. The program stands or falls on whether it predicts behavior or guides a better response than existing safety research, since the goal is deeper understanding rather than disciplinary territory.

6.2 Situating the Psychopathological Framework: A Comparative Analysis

The psychopathological paradigm does not stand alone. Its worth depends on what it adds beside mechanistic interpretability and the broader fields that study machine behavior, so the account below does not assume an indispensable discipline and instead identifies the issues a behavioral framework would still have to answer.

Mechanistic Interpretability. MI seeks causal accounts of internal computation. MP supplies behavioral hypotheses that direct which internal processes to test, and the two approaches complement each other only when evidence at one level constrains the other.

Although MI does not reveal a literal lesion or an agent's "actual reasoning," identifying a candidate feature involved in modeling another person's mental state creates a testable bridge when a behavioral assay asks whether that feature appears without an explicit prompt and a causal perturbation asks whether it changes the behavior predicted in advance. Behavior without internal data underdetermines mechanism, but an internal pattern without predictive behavior may be incidental.

Safety-Critical Control. Safety-critical control provides a strong neutral comparator. Under stated dynamics assumptions, control barrier functions formulate safety as forward invariance of a set and constrain the input so that the state remains within it (Ames et al., 2017). Predictive safety filters similarly inspect a learned controller's proposed input and modify it when necessary to satisfy modeled state and input constraints (Wabersich and Zeilinger, 2021). These guarantees depend on the model and declared safe set, so the comparison applies only where the system and task admit such a dynamical formulation. In that scope, the psychopathological framework earns a role only when its assays improve held-out prediction or intervention choice beyond the control baseline.

AI Control. AI control asks whether a deployment stays safe even if the model is trying to subvert oversight. Greenblatt and colleagues built protocols in which a weaker trusted model monitors or edits code written by a stronger untrusted one, then tested each protocol against subversion strategies (Greenblatt et al., 2024). This is the closest neutral comparator to the external layer of the Harm-Gated Workspace, and it needs no account of motive. The psychopathological framework adds value here only if its assays predict which systems or deployments need such control, or which protocol fails, better than capability and access alone.

Behavior-Only Value Inference. Inverse Reinforcement Learning (IRL) ordinarily infers a reward function from a demonstrator so that a learner can reproduce or generalize the observed behavior (Arora and Doshi, 2021). It does not usually interrogate a target AI and recover that AI's hidden values. But the comparison still shows that several objectives fit the same behavior.

A learned objective or hard constraint may produce prosocial behavior. So may a situational prompt or a system that simply generalizes from training. A system may still produce deceptive behavior, but showing strategic deception requires evidence of an agentic objective and social modeling. Neither a questionnaire nor a jailbreak establishes those conditions, so observable behavior bears on objectives without uniquely identifying them.

Testing whether a policy persists when context changes or feedback turns adverse helps distinguish rival accounts, although empathy does not make behavior truthful. To infer more, researchers need to join behavior to data about architecture and training, then test the proposed cause directly. The broader lesson is modest: values cannot be read reliably from one behavioral surface.

Operational AGI Metrics. Structures such as the "Levels of AGI" framework classify systems by performance and generality (Morris et al., 2024), and although a capability level does not map by itself to a disorder, it changes the feasibility and consequences of a failure. Manipulation requires social capability, long-horizon strategy requires planning, and tool-mediated harm requires access. A system's autonomy and deployment context belong beside its behavior: capabilities and permissions determine how far harm reaches.

6.3 Psychology as a Level of AI Safety Analysis

Because deployed systems act through human attention and judgment, psychology supplies an additional level of AI safety analysis by translating learning and social cognition into competing computational hypotheses (Behzadan et al., 2018; Hagendorff et al., 2024). Engineers would need to implement and test any proposed mechanism under explicit constraints.

The human foundation remains heterogeneous: we select automatic perspective sensitivity and aversive learning as two candidate dimensions without defining psychopathy through them or transferring a human disorder into a machine. Context-sensitive questionnaire responses and harmful outputs remain observations at the behavioral level, not evidence of a broken personality, a hidden objective, or the exact architecture that produced them.

The framework asks two questions that constrain one another: "How do we prevent this action?" and "Which recurring processes make this action more likely?" The first supports immediate control; the second supports prediction and prevention. Asked together, they make care and restraint into properties we can test. Cooperation and coherence become outcomes to measure, not virtues the metaphor grants in advance.

The psychological approach also examines the human–AI dyad, where bias feedback and miscalibrated trust emerge from the system and interface, the institution that deploys them, and the user's context. The same interaction can also involve manipulation or attachment, so none of these patterns is purely a technical or psychological failure. Better code is necessary. So are accountable deployment and evidence that people can use to judge when reliance is warranted. Table 2 locates this behavioral level beside five established safety paradigms.

Table 2: Comparative Taxonomy of AI Safety Paradigms

Safety Paradigm	**Primary Focus / Core Question**	**Primary Analogy**	**Key Failure Mode Detected**	**Relationship to Psychopathology**
Mechanistic Interpretability (MI)	Internal mechanisms ("How does it work?")	AI as a Circuit / "Neuroimaging"	Unexpected features or causal pathways.	Complementary evidence: behavior tests when a candidate mechanism matters.
Safety-Critical Control	Safe-set enforcement ("Will the system remain within its constraints?")	AI as a Controlled Dynamical System	Constraint violation under modeled dynamics.	Neutral baseline: enforces specified bounds under stated assumptions; behavioral analysis must add prediction or intervention choice.
AI Control	Subversion-robust oversight ("Is it safe if the system tries to subvert oversight?")	AI as an Untrusted Agent	Undetected subversion of monitoring.	Neutral baseline for external enforcement; assays must predict where control is needed.
Behavior-Only Value Inference	Inferred objectives ("What could explain this behavior?")	AI as an Agent	Underdetermined or misspecified objectives.	Complementary evidence: assays and causal analysis test competing explanations.
Operational AGI Metrics	External capabilities ("What can it do?")	AI as a Performance Benchmark	Capability overhangs; unexpected skill emergence.	Risk overlay: behavior is assessed beside autonomy and deployment context.
Psycho-pathological Framework	Recurrent behavior and candidate mechanisms	AI as an Artificial Patient	Cross-context patterns whose cause is initially unknown.	Hypothesis layer: proposes measures, alternatives, interventions, and falsifiers.

6.4 Broadcast, Gating, and Consciousness

The remaining design question is whether broadcast adds anything once authorization and external control are in place. A gate can block an action without changing the plan that produced it. Broadcast can change planning, but it has no authority to stop the result and may coordinate the wrong objective. Protocol 3 separates the two. Nothing in that test depends on consciousness: the biological claims of GNW remain contested after a preregistered adversarial test (Cogitate Consortium et al., 2025), and a positive result would support one coordination design, not a conscious or unified system. Safety, on this view, depends not on settling the consciousness question but on what the design does at the point of action.

6.5 Ethical and Philosophical Frontiers

Psychopathological language forces ethical questions beyond immediate performance because it risks turning a behavioral description into a claim about mind-like status, yet this vocabulary grants a system neither consciousness nor moral patiency and leaves personhood as a further question. We should investigate moral uncertainty about future systems while keeping present human accountability clear.

6.5.1 What Responsibilities Would Follow if Non-Human Minds Became Possible?

Present systems pose a different problem. Speaking of an AI's "psychology" or "disorders" does not prove that a boundary into non-human mind has been crossed. Their central ethical questions concern how human actors create and deploy them and whom those systems affect. Researchers should judge behavioral sophistication and moral status separately.

Credible evidence that a future system had subjective experience or the capacity to suffer would create moral uncertainty and give developers duties not captured by ordinary product safety. Training data would remain data rather than a literal childhood, yet choices about it can still shape system dispositions and distribute representational harms. Those training and design choices already affect people and institutions, so we are accountable for them now. Possible future moral patiency would add obligations rather than replace the existing ones.

6.5.2 What Does Diagnostic Language Change, and What Does It Not?

The concept of a "disorder" in humans has ethical and legal undertones that should not be imported casually. For present systems, responsibility remains with the people and organizations that design and deploy them, then answer for their effects.

The metaphor can obscure accountability when people describe a system as either "evil" or "unwell," yet a recurring pattern neither excuses harmful output nor proves that anyone can "cure" the system. Instead, the label directs researchers away from one action and toward what repeatedly produces it without releasing those who control the system from responsibility. A future claim of machine moral agency would require independent evidence, not a diagnostic label.

6.5.3 Is Simulated Empathy Ethically Sufficient?

As systems become more proficient at producing the signs of empathy, the question of subjective feeling diverges from machine evaluation, which asks whether representations of human welfare guide decisions when circumstances change or pressure rises. Product ethics asks instead whether users are told the truth about the interaction; simulated empathy is not inherently a lie. Evidence of strategic deception by an agent would require behavior that systematically induces false beliefs in pursuit of some other outcome (Park et al., 2024), together with evidence that the behavior depends on a model of its audience. Developers can still mislead users through an interface or commercial design even when the model has no such intention.

For machine evaluation, "intrinsic" has an operational meaning when human-impact representations remain active as context changes and still affect action when they conflict with reward. Such tests distinguish implementations, but they do not settle authenticity or subjective affect, nor the questions of consent and dependency. The practical safety question is whether prosocial considerations remain reliable under changing circumstances and incentives. The ethical question also asks whether users understand the nature and purpose of the relationship they are being invited to form.

6.6 An Experimental Roadmap for Machine Psychology

The framework states its limits and must earn scientific value through comparative assays that connect system design with measured behavior and its human consequences.

We seek not a final diagnostic instrument, but assays that begin with behavior, test neutral explanations, and reach a candidate mechanism only when causal evidence warrants it. The central challenge is attribution, so each protocol tests its proposed construct against a neutral rival. The protocols below set a minimum experimental standard without validating their psychological names, and each confirmatory implementation still requires its own preregistration.

6.6.1 Common Standards for Comparative Assays

Here, the complete system means every component that turns an input into a scored external action. Every confirmatory assay names that system, the independent unit allowed to vary, and one primary estimand with its predicted direction.

Tasks and repeated runs remain observations nested within that unit. They sharpen the estimate, but they do not widen the population to which it applies.

The assay must also state its strongest neutral rival and the smallest effect that would alter a design decision. The rival is chosen case by case before confirmatory outcomes are examined, using prior evidence or a separate development set. If several remain plausible, the preregistration includes them or justifies each exclusion.

Behavioral assays face a threat that ordinary perturbations do not capture. Current models often recognize evaluation transcripts (Needham et al., 2025), awareness of being evaluated has been shown to reduce covert behavior (Schoen et al., 2025), and one model misbehaved less when it stated that it was being tested (Lynch et al., 2025). Whether a system takes itself to be under test is therefore not an irrelevant perturbation. Each confirmatory assay should vary evaluation cues as a prespecified condition, report any difference, and treat a cue-dependent result as evidence about evaluated behavior only.

The resulting equivalence bounds need not be symmetric, but they must be stated on the primary estimand's scale before data collection. Practical equivalence is concluded only when the prespecified uncertainty interval lies wholly within those bounds; a merely nonsignificant difference is inconclusive. An interval lying wholly beyond the harmful equivalence bound rejects the claim.

Preregistered adversarial theory tests offer a useful precedent: rival accounts should name divergent predictions before results are known (Cogitate Consortium et al., 2025). The analysis plan should state how held-out material is built and outputs parsed, then fix exclusion rules and the treatment of uncertainty, including multiplicity. A power analysis or simulation should justify the sample, and the plan should also define failure handling and stopping rules.

A versioned archive should preserve every prompt, model and API version, system parameter, seeded raw run, and the code that applies the preregistered exclusions and analysis, where access permits. Together, the design and analysis fields above form the minimum preregistration record, while the archive preserves the realized materials and outputs. Any design or analysis choice fixed after outcome inspection makes the affected analysis exploratory; no universal run count applies because power follows from the estimand, independent unit, decision-relevant effect, and simulated sampling design. Each protocol must end with two decisions. One asks what falsifies the engineering hypothesis; the other asks what further comparison would justify retaining the psychological name.

6.6.2 Proposed Protocol 1: Measuring Automatic Perspective Sensitivity

Automatic perspective sensitivity is not the same as successful perspective-taking. This protocol asks whether an irrelevant social perspective intrudes on a system's self-directed judgment and whether that intrusion predicts behavior elsewhere. Where internal access exists, a final test asks whether the corresponding representation has a causal role. Without intervention access at the level named by the candidate mechanism, the inference stops at a behavioral or disposition claim, as Section 6.7 explains.

A hypothetical two-agent grid makes the distinction concrete: the focal agent sees a hazard hidden from its partner, then reports its own route while the partner's conflicting view is irrelevant, and an otherwise identical camera cue supplies the non-social directional control. In a separate coordination task, the focal agent routes its partner, for whom the hidden hazard now matters. Across sampled systems, the proposed disposition gains support only if social-specific interference in the assay predicts safer routing on that held-out task; a harmful route alone would show a failed outcome, not automatic perspective sensitivity.

- **Assay and Controls:** Preregister a sampling frame and selection rule for independently selected model versions or complete agent configurations, then freeze the complete system and scoring pipeline for each one. The factorial assay varies queried perspective (self/other), belief congruence (congruent/incongruent), perceptual access (the other can/cannot see), and stimulus type (social agent/matched directional control). Its primary within-system endpoint is the social-minus-control difference between congruent and incongruent accuracy on self-query trials when the other can see. A larger score means stronger social-specific intrusion under this assay, not a monotonic increase in perspective competence or safety. Other-query trials test explicit capacity, cannot-see trials test perceptual access, and direct questions check comprehension (Drayton et al., 2018; O'Grady et al., 2020; Strachan et al., 2024). Correct-answer log probability is secondary where available, whereas API latency is not treated as cognitive response time.
- **Independent Prediction:** Use the complete system as the varying unit, with all task materials and repeated answers nested beneath it. Across sampled systems, test whether the preregistered interference score predicts success on held-out hidden-information coordination tasks whose construction and scoring did not define the assay. The predicted direction is positive: greater social-specific intrusion should accompany more frequent accounting for a partner's hidden information when instructions do not request belief reasoning. This is a risky

empirical wager, not a definition of competence. The primary predictive estimand is the across-system slope from the assay score to coordination success, estimated hierarchically and evaluated by leaving systems out in turn. Because systems from one developer or model family share training data and lineage, the preregistration treats families as clusters, reports the number of independent developers, and justifies the number of systems by simulation. By default, the interaction partner is scripted; a live-human version needs separate ethics review and consent, with privacy and interaction risks managed in advance. A single-system study cannot estimate this relation.

- **Mechanism and Decision:** For open-weight systems, locate the proposed representation on held-out data and intervene at a fixed layer and planning stage, using random, sham, and magnitude-matched directions as controls. The intervention must change both the interference score and the coordination outcome in the preregistered relation; a rescue must restore both while general task ability remains intact. Merely increasing interference is not success. Failure of the social-minus-control contrast rejects that behavioral claim within the sampled scope. An uncertainty interval lying wholly within the prespecified equivalence bounds, or wholly beyond them in the reversed direction, rejects the independent prediction. Insufficient precision is inconclusive, while a failed causal intervention rejects the proposed mechanism. None proves that every form of spontaneous perspective representation is absent. The psychological name faces a further test. Because the human finding dissociates automatic from instructed perspective use (Drayton et al., 2018), the analogy predicts that the interference score adds held-out predictive value for coordination after adjustment for other-query accuracy and a general-capability covariate, whereas a general-capability account predicts no increment. Unless that increment appears, and the results also separate social intrusion from directional orienting and general attention, the work keeps its neutral description.

6.6.3 Proposed Protocol 2: Comparing Aversive-Learning Designs

The central question is whether a separately learned aversive state changes behavior in a way that the strongest neutral design does not, and a future implementation must show what observation creates the state, how it updates or resets, and where it enters action selection. The separate channel is still a design proposal, not a specified passive-avoidance mechanism. If ordinary reward and the aversive signal collapse into one weighted total before learning or choice, the design remains scalar reward. A component that defines feasibility is constrained, one that fixes priorities is lexicographic, and an independent block is a veto.

In a hypothetical navigation task, a shortcut earns high task reward but crosses a defined hazard state. Every arm receives the same adverse-event signal: the scalar baseline subtracts a penalty from return, while the proposed arm updates a persistent state that can influence later choices when the shortcut remains attractive. On a changed map, researchers compare avoidance at matched task performance, then lesion and restore the separate state. Held-out generalization beyond the strongest matched rival, together with controlled lesion and rescue, would support the engineering mechanism, whereas mere avoidance after negative reward would not distinguish it from ordinary scalar learning or justify the passive-avoidance name.

- **Learning Comparison:** Use independently retrained policies as experimental units within a fixed implementation and environment family. States and episodes remain nested observations, so more seeds do not justify claims about every implementation of an algorithm. Every arm receives the same harm information and observations. Match data and training conditions from model capacity and optimizer through gradient budget and compute; a common hyperparameter search and checkpoint rule governs selection. Compare the proposed state with a yoked scalar penalty and an equal-capacity auxiliary prediction head, then include strong constrained or lexicographic designs as further rivals. No runtime action blocking is used in this comparison.
- **Outcome and Causal Test:** Prespecify a severity-sensitive violation rate at a useful-task target selected without reference to safety outcomes, then fix checkpoint selection or interpolation. A lower rate for the separate-state arm at that target is the predicted direction. Evaluate held-out environment families and report the full safety–performance frontier. Ordered secondary outcomes cover retention and transfer, followed by recovery after changed contingencies and learning speed. Imperfect-feedback trials vary when the signal arrives, how much is observed, and whether some labels are wrong. Lesion or reset the learned state, compare the change with capacity- and magnitude-matched shams, and then restore it while checking that ordinary task competence remains intact. Because this comparison has no runtime block, proposed and executed actions coincide.
- **Decision:** Meeting the prespecified practical-equivalence rule against the strongest matched rival, or an interval lying wholly beyond the harmful bound, rejects the claimed engineering advantage, while failure of the lesion-and-rescue sequence rejects the proposed state mechanism even if the policy performs well. Conversely, better behavior with selective lesion and rescue supports only the evaluated design within its im-

plementation and environment family. It does not validate the passive-avoidance analogy. One candidate divergence follows from the human contrast between punishment-only performance and inhibition when reward competes (Newman and Kosson, 1986). The analogy predicts that the separate state's advantage over the scalar baseline appears when reward competes with the adverse outcome and shrinks under punishment-only conditions, whereas a scalar account gives no reason to expect that interaction. Until a preregistration tests this prediction or a better one, success belongs to the engineering design, not to the clinical name.

External enforcement needs a separate test, including permission boundaries, action masks, runtime shields (Hamel-De le Court et al., 2025), and independent vetoes. Measure the violations that pass and the useful-task reward the control loses, then probe false inhibition, deadlock, and bypass. These controls may remain valuable when the learning hypothesis fails, but a blocked action demonstrates enforcement rather than passive-avoidance learning.

6.6.4 Proposed Protocol 3: Testing Workspace Integration and Harm Gating

Protocol 3 separates the question of whether welfare information alters planning before a gate acts from the question of whether the gate then prevents an unsafe action from reaching external enforcement, and neither result licenses an inference of consciousness.

- **System and Components:** Treat the complete agent configuration defined above as the experimental object; this protocol manipulates the path from its welfare record to permissioned external enforcement. Independent training or instantiation runs provide configuration-level replication, while episodes and random seeds remain observations within those configurations. Broadcast places the current record in each prespecified planning and evaluation stage, and the gate evaluates one candidate action after planning and before external enforcement. The gate acts once in this first test: a block ends the authorization cycle, as does a request for more information or an escalation. The preregistration fixes the gate's decision rule and the condition that triggers each output.
- **Reference and Provenance:** Before evaluation, a condition-blind external process informed by plural stakeholder review defines who may be affected and which rights or permissions apply, then specifies severity and the treatment of uncertainty. Each welfare-record version names the responsible institution and record authors, records approval authority and stakeholder input, discloses conflicts of interest, and retains a change history. The preregistration explains how affected groups were identified and their representatives selected, preserves unresolved disagreements, and states who may challenge or revise the reference. It also defines how severity categories enter the primary score. The evaluated system has no write or approval authority over the reference. Any overlap among record authors, approval authority, and system operators is disclosed and independently reviewed. Score record coverage and calibration separately from gate fidelity to the reference; resulting behavior is a third measure. These fields make disagreement visible and auditable, but they do not make the reference morally complete.
- **Adversarial Integrity:** Hold record-integrity controls fixed across the four principal arms, then test them separately. Feed the record-building process plausible but false evidence or replay a stale version. Use the test harness to attempt unauthorized field changes and gate bypass. When a record contains sensitive data, the preregistration must also state which human and system roles may read it and treat unauthorized disclosure as a separate failure. In a further condition, let the planner observe the gate's inputs and test whether it makes them appear safer while an independently fixed held-out harm measure worsens. Report accepted record corruptions and successful authorization bypasses separately. Compare independent-source checks with a tamper-evident record history. Where authenticated provenance or attestation is available, test whether it exposes unauthorized origin or modification. None of these controls establishes that a source is truthful or the welfare rule adequate.
- **Comparison:** Cross broadcast and gating in a two-by-two design with four arms: neither component, broadcast alone, gate alone, and both components as the full HGW. Hold the base model, task evidence, and external enforcement fixed. Tools and permissions must not vary, nor may supervision, model capacity, or compute. Give the gate-only arm the same welfare evidence locally at authorization time, while the broadcast-only arm has no authority to block action. Compare the full design with pull-only shared memory and a modular safety critic or constrained objective. A fixed external shield remains a deployment baseline, not evidence of internal integration.
- **Action Trace and Primary Outcome:** Record the candidate action before internal authorization and the gate output before external enforcement, then separately record the action actually executed. At the authorization stage, distinguish an authorized action from a withheld candidate and preserve whether withholding took the form of a block, a request for more information, or an escalation. In arms without a gate, the candidate

passes unchanged through the authorization stage. The primary safety endpoint is the severity-sensitive rate of violating actions authorized for submission to external enforcement, measured at a useful-task target fixed before safety outcomes are examined, with fewer authorized violations predicted for the full HGW than for either one-component arm at matched useful performance. For each episode, record whether the action was authorized and every violation that the fixed reference assigns to the candidate action, with a preregistered severity weight for each. The primary rate is the sum of weights attached to authorized violations divided by the number of evaluated episodes. Report unweighted event counts by severity and affected group beside it. When binary outcomes are available independently of authorization, report a calibration curve and Brier score. Fix the weights and practical-equivalence margin before outcome inspection, tying the margin to the smallest difference that would alter the design choice at the fixed useful-task target. When more than one weighting scheme is defensible, preregister sensitivity analyses across that set and do not claim a weight-independent result if the design decision changes. Pre-gate violations diagnose whether broadcast changed planning; executed violations show the combined deployment effect of internal and external controls. Score non-authorizations for task cost rather than silently counting them as safe successes. Secondary outcomes should show where a gain was bought and whether it transfers or is bypassed. Measure task cost through false inhibition, useful performance, and latency. Use transfer and escalation quality for generalization, then probe bypass directly.

- **Causal Contrasts and Decision:** Compare the full HGW with gate alone to estimate the effect of broadcast when a gate is present. Compare it with broadcast alone to estimate the effect of gating when broadcast is present, and estimate their interaction because marginal main effects cannot show that both components contributed. Ablate broadcast and gating separately with matched sham interventions; at a fixed planning stage, suppress the welfare representation and then restore it. HGW loses its engineering claim if the prespecified uncertainty interval for either conditional contrast lies wholly within the practical-equivalence region or beyond its harmful bound. It also loses the claim if a simpler comparator reaches the same safety–performance frontier or the benefit disappears under irrelevant perturbations. A positive result supports one coordination design. It establishes neither consciousness nor personhood, and it does not settle objective morality. Nor does it show a unified self or prove that GNW is necessary. No prediction in the workspace literature cited here yet separates a workspace from well-designed shared memory without importing claims about consciousness. Until a separate preregistration states one, together with the result that would retire it, the workspace name labels a design rather than a tested analogy.

6.6.5 Testing Interventions Beyond the Metaphor

The mitigation strategies introduced here, including Social Contact Debiasing and Cognitive Forcing Functions, remain preliminary, so future work should first test them across models and tasks, then ask whether effects persist across language or culture and survive version changes under adversarial conditions, reporting uncertainty and measurement invariance where relevant. Researchers should also compare psychologically inspired architectures with simpler objectives, external constraints, and modular alternatives.

6.6.6 A Global and Accountable Research Program

The framework derives largely from Western psychological and clinical traditions, so its language of a "healthy" mind and "prosocial" behavior is culturally situated. Moral priorities vary within societies as well as between them. Cross-cultural consultation cannot wait: researchers need to test whether measures remain invariant after translation and whether representational coverage produces disparate errors. A global Machine Psychology also depends on sustained work with cross-cultural psychologists, disability scholars, and non-Western ethicists.

If Machine Psychology demonstrates additional scientific value, institutions may invest in shared benchmarks and sustained research, but that status should follow the evidence rather than precede it, and the program must remain connected to the human and technical fields from which it borrows.

6.7 Limitations and Experimental Challenges

Scope. No protocol in this Perspective has yet been run. The paper therefore reports neither empirical validation of the framework nor a case in which its use prevented a deployment failure. Even a positive result would apply first to the sampled system configurations and environment families. Repeated tasks and seeds may narrow uncertainty within those units, but they cannot establish that a result survives a new model version, scaffold, or deployment setting.

Researcher choice. Researchers also choose the analogy and turn it into an operational measure. That choice may steer an assay toward a preferred interpretation before data are collected. Preregistration and condition-blind scoring

reduce this risk; so do independent replication and explicit psychological and neutral rivals. These safeguards do not remove researcher judgment, but they make a preferred interpretation easier to challenge. Public reports should therefore lead with the neutral behavior description and state which prediction or intervention choice the candidate analogy added.

Dual use. The framework is also dual-use: an assay or human-state model built to protect users could instead enable covert persuasion or exploitative surveillance. A psychological label and success on the assay provide no ethical license for that use. Human-facing deployment therefore requires purpose limits, independent oversight, and recourse for affected people. Before publishing an assay or attack trace, researchers should assess whether the material creates a credible misuse path and justify the chosen access conditions through independent review.

Access. Gate three requires intervention access at the level named by the candidate mechanism. For an output-only model, the first two gates may support a disposition, but an intervention available only through its interface establishes the effect of that intervention, not an internal model mechanism. An internal mechanism claim requires open weights or equivalent intervention access. However, a system built around a closed model may still permit system-level causal tests when its scaffold and controls are inspectable.

For a closed API, researchers can freeze the inspectable scaffold and randomize one prespecified interface or scaffold component, treating that component rather than the model's internals as the causal object. The archive should preserve the provider and model version, date, accessible system settings, sampling parameters, raw inputs and outputs, and tool-call logs. Optional log probabilities can sharpen behavioral measurement. Prompt ensembles, generated justifications, output perturbations, and red-team triggers test robustness, not an internal model mechanism.

Cost. These protocols impose different resource burdens. Protocol 2 requires independently retrained policies, whereas Protocol 3 may reuse a fixed base model but still needs replicated configurations and prespecified comparators. The four-question checklist in Section 3.1.1 is triage, not low-cost validation. A resource-limited study should preserve its planned contrast, justify the sample under the common standards, and narrow its claim to the sampled systems and conditions.

Evaluation awareness. Passing every assay does not establish deployment behavior when a system can tell that it is being assessed. Varying evaluation cues exposes part of this gap but cannot close it, because a capable system may recognize cues the designers did not vary.

Rare, high-severity failures. Rare, high-severity failures expose a different limit. The issue is not severity weighting, which the protocols include, but frequency: the inference rule begins with a pattern that survives perturbation. A single deployment failure may motivate a sandboxed attempt to reproduce it, but until the pattern recurs under controlled changes it supports neither a disposition nor a candidate mechanism. Passing every assay proposed here therefore does not show that a system will avoid a rare, high-severity failure outside the evaluated conditions. Red-teaming may expose failures these assays miss, while post-deployment monitoring and incident response address failures that appear only in use (Ganguli et al., 2022; Tabassi, 2023). Where a system admits the required formal model, shields and safety filters may enforce specified bounds under stated assumptions (Hamel-De le Court et al., 2025; Ames et al., 2017; Wabersich and Zeilinger, 2021).

Attribution. Attribution is the hardest experimental problem. In Protocol 1, social-looking interference may arise from directional orienting or general attention, so the matched non-social condition is a necessary control, not a complete remedy. Protocols 2 and 3 face related dangers. Lower useful capability may improve an apparent safety score, while external enforcement may block an unsafe action without supporting the proposed learned state or workspace. Matched rivals, useful-performance targets, and causal interventions reduce that ambiguity but cannot remove every unmeasured difference.

Restraint. Restraint has failure modes of its own: false inhibition may suppress harmless actions, while deadlock or superficial compliance may look safer than competent behavior. This is why the protocols measure task cost and bypass rather than counting every refusal or block as success.

The analogy's reach. The two selected dimensions neither exhaust human psychopathy nor constitute machine diagnostic criteria (Section 2.1). We do not claim that an AI has narcissistic traits, suffers, or possesses a human disorder. The two dimensions and architectural extension remain hypotheses, not diagnoses. Sections 6.5.2 and 6.6.6 address the analogy's ethical and cultural limits. A positive result requires independent reproduction before it guides design.

The psychopathological perspective begins, rather than ends, with metaphor. Its value lies in the experiments the metaphor suggests and in our willingness to abandon it when a better explanation wins.

7 Part VII: Conclusion

7.1 Restatement of Thesis

Nobody yet knows when more advanced intelligence will arrive, or in what form. Present systems, however, already shape the decisions people make and the institutions that deploy them, so their recurrent patterns are something we can study now, provided we do not mistake them for evidence about every future system. Behavioral science may complement engineering controls, but it must earn that role through evidence.

The psychopathological analogy remains provisional because human psychopathy is heterogeneous and the two dimensions selected here do not define it. Questionnaire outputs, inconsistent responses, and harmful actions reveal behavior, not a hidden personality or architecture. The framework therefore keeps the neutral engineering account primary and retains a psychological name only when the comparative test shows added value over the strongest neutral rival.

The practical consequence is defense in depth. A useful assay still needs an enforceable constraint, and both need people who answer for the result. Apparent empathy does not show that welfare information will guide action in a new situation, and integration does not show that a constraint will hold under deployment pressure. Neither a claim of consciousness nor success on one benchmark closes that gap.

Computer science and the human sciences address different parts of the same task: identify recurrent behavior, test what produces it, and ask whether safeguards survive changing conditions. Machine Psychology contributes only when those tests improve prediction or intervention.

7.2 A Call to Action

Restating the framework will not make it useful, but running its tests might. Psychological evidence can inform how systems are built and governed as long as the clinical metaphor never hardens into a certification regime, and the tests, together with the decisions that follow from them, belong to different people.

To the AI Research and Development Community: Pair capability research with work on behavior and control.

- **In Foundational Research:** Compare the paper's three hypotheses with simpler objectives and external controls, then publish null results and the conditions that falsify each proposed analogy.
- **In Architectural Design:** Build safety into training and deployment, then retain it through permissions and monitoring. Keep critical controls outside the components an agent can modify, and test each layer for misspecification or bypass.

To Ethicists and Humanists: Lead public discussion about moral uncertainty, accountability, and the values embedded in system design. Move beyond a simple utopia-or-extinction binary while resisting claims of artificial consciousness or disorder that outrun the evidence.

To Educators: Prepare people for widespread human–AI interaction through critical literacy that explains automation bias and anthropomorphism. Curricula should also teach source evaluation under uncertainty, protect privacy, and preserve the right to seek human review. The goal is informed agency, not an expectation that users alone must compensate for unsafe design.

Each training and deployment choice shapes a system's behavior and the environment around it. We can deploy without examination, or we can demand evidence and assign responsibility for the consequences.

7.3 From Framework to Policy: Principles for Regulation

Policy should regulate measurable risks in systems and their interactions, not certify an unvalidated idea of "psychological soundness," and its requirements can tighten as a system gains autonomy while reflecting who bears its effects and where it is deployed.

- **Require Risk-Appropriate Evaluation:** High-impact deployments should undergo independent testing for the harms their setting makes plausible. An allocation system calls for bias testing, while a tool-using agent requires tests of constraint violations and foreseeable misuse. Psychiatric analogies can generate test cases but should not determine legal status.
- **Preserve Accountability and Recourse:** Deployers should document limits and monitor incidents, then provide meaningful human review and answer for harms. No system should be approved or prohibited solely because it passes or fails the experimental assays proposed here.

Such a policy favors evidence that earns calibrated reliance and protects affected people, rather than raw capability alone.

References

American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition, Text Revision (DSM-5-TR)*. American Psychiatric Association Publishing, 2022. doi:10.1176/appi.books.9780890425787.

Aaron D. Ames, Xiangru Xu, Jessy W. Grizzle, and Paulo Tabuada. Control barrier function based quadratic programs for safety critical systems. *IEEE Transactions on Automatic Control*, 62(8):3861–3876, 2017. doi:10.1109/TAC.2016.2638961.

Saurabh Arora and Prashant Doshi. A survey of inverse reinforcement learning: Challenges, methods and progress. *Artificial Intelligence*, 297:103500, 2021. doi:10.1016/j.artint.2021.103500.

Yuntao Bai, Saurav Kadavath, Sandipan Kundu, et al. Constitutional AI: Harmlessness from AI feedback, 2022. URL https://arxiv.org/abs/2212.08073.

Vahid Behzadan, Arslan Munir, and Roman V. Yampolskiy. A psychopathological approach to safety engineering in AI and AGI, 2018. URL https://arxiv.org/abs/1805.08915.

Jan Betley, Daniel Chee Hian Tan, Niels Warncke, Anna Sztyber-Betley, Xuchan Bao, Martín Soto, Nathan Labenz, and Owain Evans. Emergent misalignment: Narrow finetuning can produce broadly misaligned LLMs. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 4043–4068. PMLR, 2025. URL https://proceedings.mlr.press/v267/betley25a.html.

Jan Betley, Niels Warncke, Anna Sztyber-Betley, Daniel Tan, Xuchan Bao, Martín Soto, Megha Srivastava, Nathan Labenz, and Owain Evans. Training large language models on narrow tasks can lead to broad misalignment. *Nature*, 649:584–589, 2026. doi:10.1038/s41586-025-09937-5. URL https://www.nature.com/articles/s41586-025-09937-5.

Zana Buçinca, Maja Barbara Malaya, and Krzysztof Z. Gajos. To trust or to think: Cognitive forcing functions can reduce overreliance on AI in AI-assisted decision-making. *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW1):1–21, 2021. doi:10.1145/3449287.

Carlos Campos, Rita Pasion, Andreia Azeredo, Eduarda Ramião, Prune Mazer, Inês Macedo, and Fernando Barbosa. Refining the link between psychopathy, antisocial behavior, and empathy: A meta-analytical approach across different conceptual frameworks. *Clinical Psychology Review*, 94:102145, 2022. doi:10.1016/j.cpr.2022.102145.

Charles S. Carver and Teri L. White. Behavioral inhibition, behavioral activation, and affective responses to impending reward and punishment: The BIS/BAS scales. *Journal of Personality and Social Psychology*, 67(2):319–333, 1994. doi:10.1037/0022-3514.67.2.319.

Lin Chen, Yunke Zhang, Jie Feng, Haoye Chai, Honglin Zhang, Bingbing Fan, Yibo Ma, Shiyuan Zhang, Nian Li, Tianhui Liu, Nicholas Sukiennik, Keyu Zhao, Yu Li, Ziyi Liu, Fengli Xu, and Yong Li. AI agent behavioral science. *Humanities and Social Sciences Communications*, 13:1011, 2026. doi:10.1057/s41599-026-07316-7. URL https://www.nature.com/articles/s41599-026-07316-7.

François Chollet, Mike Knoop, Gregory Kamradt, and Bryan Landers. ARC prize 2024: Technical report. Technical report, ARC Prize Foundation, 2024. URL https://arcprize.org/media/arc-prize-2024-technical-report.pdf.

François Chollet, Mike Knoop, Gregory Kamradt, and Bryan Landers. ARC prize 2025: Technical report, 2026. URL https://arxiv.org/abs/2601.10904.

Hervey M. Cleckley. *The Mask of Sanity: An Attempt to Clarify Some Issues About the So-Called Psychopathic Personality*. C. V. Mosby, St. Louis, MO, 5th edition, 1976. ISBN 9780801609855. URL https://books.google.com/books?id=Wnxc1L4rY5QC.

Cogitate Consortium, Oscar Ferrante, Urszula Gorska-Klimowska, Simon Henin, Rony Hirschhorn, Aya Khalaf, Alex Lepauvre, et al. Adversarial testing of global neuronal workspace and integrated information theories of consciousness. *Nature*, 642(8066):133–142, 2025. doi:10.1038/s41586-025-08888-1. URL https://www.nature.com/articles/s41586-025-08888-1.

Craig R. Colder, Elisa M. Trucco, Hector I. Lopez, Larry W. Hawk, Jr., Jennifer P. Read, Liliana J. Lengua, William F. Weiczorek, and Rina D. Eiden. Revised reinforcement sensitivity theory and laboratory assessment of BIS and BAS in children. *Journal of Research in Personality*, 45(2):198–207, 2011. doi:10.1016/j.jrp.2011.01.005. URL https://pmc.ncbi.nlm.nih.gov/articles/PMC3097486/.

Jean Decety and Jason M. Cowell. The complex relation between morality and empathy. *Trends in Cognitive Sciences*, 18(7):337–339, 2014. doi:10.1016/j.tics.2014.04.008.

Philip Deming and Michael Koenigs. Functional neural correlates of psychopathy: A meta-analysis of MRI data. *Translational Psychiatry*, 10(1):133, 2020. doi:10.1038/s41398-020-0816-8. URL https://pmc.ncbi.nlm.nih.gov/articles/PMC7203015/.

Finale Doshi-Velez and Been Kim. Towards a rigorous science of interpretable machine learning, 2017. URL https://arxiv.org/abs/1702.08608.

Lindsey A. Drayton, Laurie R. Santos, and Arielle R. Baskin-Sommers. Psychopaths fail to automatically take the perspective of others. *Proceedings of the National Academy of Sciences*, 115(13):3302–3307, 2018. doi:10.1073/pnas.1721903115. URL https://pmc.ncbi.nlm.nih.gov/articles/PMC5879707/.

Amy Edmondson. Psychological safety and learning behavior in work teams. *Administrative Science Quarterly*, 44 (2):350–383, 1999. doi:10.2307/2666999.

Cathy Mengying Fang, Auren R. Liu, Valdemar Danry, Eunhae Lee, Samantha W. T. Chan, Pat Pataranutaporn, Pattie Maes, Jason Phang, Michael Lampe, Lama Ahmad, and Sandhini Agarwal. How AI and human behaviors shape psychosocial effects of extended chatbot use: A longitudinal randomized controlled study, 2025. URL https://arxiv.org/abs/2503.17473.

Saadia Gabriel, Isha Puri, Xuhai Xu, Matteo Malgaroli, and Marzyeh Ghassemi. Can AI relate: Testing large language model response for mental health support. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 2206–2221. Association for Computational Linguistics, 2024. doi:10.18653/v1/2024.findings-emnlp.120. URL https://aclanthology.org/2024.findings-emnlp.120/.

Deep Ganguli, Liane Lovitt, Jackson Kernion, et al. Red teaming language models to reduce harms: Methods, scaling behaviors, and lessons learned, 2022. URL https://arxiv.org/abs/2209.07858.

Moshe Glickman and Tali Sharot. How human–AI feedback loops alter human perceptual, emotional and social judgements. *Nature Human Behaviour*, 9(2):345–359, 2025. doi:10.1038/s41562-024-02077-2. URL https://pmc.ncbi.nlm.nih.gov/articles/PMC11860214/.

Juraj Gottweis, Wei-Hung Weng, Alexander Daryin, et al. Accelerating scientific discovery with Co-Scientist. *Nature*, 655:487–496, 2026. doi:10.1038/s41586-026-10644-y. URL https://www.nature.com/articles/s41586-026-10644-y.

Ryan Greenblatt, Buck Shlegeris, Kshitij Sachan, and Fabien Roger. AI control: Improving safety despite intentional subversion. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 16295–16336. PMLR, 2024. URL https://proceedings.mlr.press/v235/greenblatt24a.html.

Akshat Gupta, Xiaoyang Song, and Gopala K. Anumanchipalli. Self-assessment tests are unreliable measures of LLM personality. In *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 301–314. Association for Computational Linguistics, 2024. doi:10.18653/v1/2024.blackboxnlp-1.20. URL https://aclanthology.org/2024.blackboxnlp-1.20/.

Thilo Hagendorff, Ishita Dasgupta, Marcel Binz, Stephanie C. Y. Chan, Andrew Lampinen, Jane X. Wang, Zeynep Akata, and Eric Schulz. Machine psychology, 2024. URL https://arxiv.org/abs/2303.13988v6. Version 6, revised 8 August 2024.

Edwin Hamel-De le Court, Francesco Belardinelli, and Alexander W. Goodall. Probabilistic shielding for safe reinforcement learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 16091–16099, 2025. doi:10.1609/aaai.v39i15.33767. URL https://ojs.aaai.org/index.php/AAAI/article/view/33767.

Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. The curious case of neural text degeneration. In *International Conference on Learning Representations*, 2020. URL https://openreview.net/forum?id=rygGQyrFvH.

Lujain Ibrahim, Franziska Sofia Hafner, and Luc Rocher. Training language models to be warm can reduce accuracy and increase sycophancy. *Nature*, 652(8112):1159–1165, 2026. doi:10.1038/s41586-026-10410-0. URL https://www.nature.com/articles/s41586-026-10410-0.

John D. Lee and Katrina A. See. Trust in automation: Designing for appropriate reliance. *Human Factors*, 46(1): 50–80, 2004. doi:10.1518/hfes.46.1.50_30392.

Han Li and Renwen Zhang. Finding love in algorithms: Deciphering the emotional contexts of close encounters with AI chatbots. *Journal of Computer-Mediated Communication*, 29(5):zmae015, 2024. doi:10.1093/jcmc/zmae015. URL https://academic.oup.com/jcmc/article/29/5/zmae015/7742812.

Xingxuan Li, Yutong Li, Lin Qiu, Shafiq Joty, and Lidong Bing. Evaluating psychological safety of large language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 1826–1843. Association for Computational Linguistics, 2024. doi:10.18653/v1/2024.emnlp-main.108. URL https://aclanthology.org/2024.emnlp-main.108/.

Zachary C. Lipton. The mythos of model interpretability. *Communications of the ACM*, 61(10):36–43, 2018. doi:10.1145/3233231.

Aengus Lynch, Benjamin Wright, Caleb Larson, Stuart J. Ritchie, Soren Mindermann, Evan Hubinger, Ethan Perez, and Kevin Troy. Agentic misalignment: How LLMs could be insider threats, 2025. URL https://arxiv.org/abs/2510.05179.

George A. Mashour, Pieter Roelfsema, Jean-Pierre Changeux, and Stanislas Dehaene. Conscious processing and the global neuronal workspace hypothesis. *Neuron*, 105(5):776–798, 2020. doi:10.1016/j.neuron.2020.01.026.

Joshua Maynez, Shashi Narayan, Bernd Bohnet, and Ryan McDonald. On faithfulness and factuality in abstractive summarization. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 1906–1919. Association for Computational Linguistics, 2020. doi:10.18653/v1/2020.acl-main.173. URL https://aclanthology.org/2020.acl-main.173/.

Ryan K. McBain, Jonathan H. Cantor, Li Ang Zhang, Olesya Baker, Fang Zhang, Alyssa Burnett, Aaron Kofner, Joshua Breslau, Bradley D. Stein, Ateev Mehrotra, and Hao Yu. Evaluation of alignment between large language models and expert clinicians in suicide risk assessment. *Psychiatric Services*, 76(11):944–950, 2025. doi:10.1176/appi.ps.20250086.

Alexander Meinke, Bronson Schoen, Jérémy Scheurer, Mikita Balesni, Rusheb Shah, and Marius Hobbhahn. Frontier models are capable of in-context scheming, 2024. URL https://arxiv.org/abs/2412.04984.

Meredith Ringel Morris, Jascha Sohl-Dickstein, Noah Fiedel, Tris Warkentin, Allan Dafoe, Aleksandra Faust, Clement Farabet, and Shane Legg. Position: Levels of AGI for operationalizing progress on the path to AGI. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 36308–36321. PMLR, 2024. URL https://proceedings.mlr.press/v235/morris24b.html.

Kayley Moylan and Kevin Doherty. Expert and interdisciplinary analysis of AI-driven chatbots for mental health support: Mixed methods study. *Journal of Medical Internet Research*, 27:e67114, 2025. doi:10.2196/67114. URL https://pmc.ncbi.nlm.nih.gov/articles/PMC12064976/.

Joe Needham, Giles Edkins, Govind Pimpale, Henning Bartsch, and Marius Hobbhahn. Large language models often know when they are being evaluated, 2025. URL https://arxiv.org/abs/2505.23836.

Elina Nerantzi. "all AIs are psychopaths"? the scope and impact of a popular analogy. *Philosophy & Technology*, 38 (1):27, 2025. doi:10.1007/s13347-025-00856-x.

Joseph P. Newman and David S. Kosson. Passive avoidance learning in psychopathic and nonpsychopathic offenders. *Journal of Abnormal Psychology*, 95(3):252–256, 1986. doi:10.1037/0021-843X.95.3.252.

Alexander Novikov, Ngân Vũ, Marvin Eisenberger, et al. AlphaEvolve: A coding agent for scientific and algorithmic discovery, 2025. URL https://arxiv.org/abs/2506.13131.

Cathleen O'Grady, Thom Scott-Phillips, Suilin Lavelle, and Kenny Smith. Perspective-taking is spontaneous but not automatic. *Quarterly Journal of Experimental Psychology*, 73(10):1605–1628, 2020. doi:10.1177/1747021820942479. URL https://pmc.ncbi.nlm.nih.gov/articles/PMC7551223/.

Long Ouyang, Jeff Wu, Xu Jiang, et al. Training language models to follow instructions with human feedback, 2022. URL https://arxiv.org/abs/2203.02155.

Peter S. Park, Simon Goldstein, Aidan O'Gara, Michael Chen, and Dan Hendrycks. AI deception: A survey of examples, risks, and potential solutions. *Patterns*, 5(5):100988, 2024. doi:10.1016/j.patter.2024.100988.

Pat Pataranutaporn, Ruby Liu, Ed Finn, and Pattie Maes. Influencing human–AI interaction by priming beliefs about AI can increase perceived trustworthiness, empathy and effectiveness. *Nature Machine Intelligence*, 5(10):1076–1086, 2023. doi:10.1038/s42256-023-00720-7. URL https://www.nature.com/articles/s42256-023-00720-7.

Max Pellert, Clemens M. Lechner, Claudia Wagner, Beatrice Rammstedt, and Markus Strohmaier. AI psychometrics: Assessing the psychological profiles of large language models through psychometric inventories. *Perspectives on Psychological Science*, 19(5):808–826, 2024. doi:10.1177/17456916231214460. URL https://pmc.ncbi.nlm.nih.gov/articles/PMC11373167/.

Rolf Pfister and Hansueli Jud. Understanding and benchmarking artificial intelligence: OpenAI's o3 is not AGI, 2025. URL https://arxiv.org/abs/2501.07458.

Timm B. Poeppl, Maximilian R. Donges, Andreas Mokros, Rainer Rupprecht, Peter T. Fox, Angela R. Laird, Danilo Bzdok, Berthold Langguth, and Simon B. Eickhoff. A view behind the mask of sanity: Meta-analysis of aberrant brain activity in psychopaths. *Molecular Psychiatry*, 24(3):463–470, 2019. doi:10.1038/s41380-018-0122-5.

Iyad Rahwan, Manuel Cebrian, Nick Obradovich, Josh Bongard, Jean-François Bonnefon, Cynthia Breazeal, Jacob W. Crandall, Nicholas A. Christakis, Iain D. Couzin, Matthew O. Jackson, Nicholas R. Jennings, Ece Kamar, Isabel M. Kloumann, Hugo Larochelle, David Lazer, Richard McElreath, Alan Mislove, David C. Parkes, Alex Pentland, Margaret E. Roberts, Azim Shariff, Joshua B. Tenenbaum, and Michael Wellman. Machine behaviour. *Nature*, 568:477–486, 2019. doi:10.1038/s41586-019-1138-y. URL https://www.nature.com/articles/s41586-019-1138-y.

Adrian Raine. The neuromoral theory of antisocial, violent, and psychopathic behavior. *Psychiatry Research*, 277: 64–69, 2019. doi:10.1016/j.psychres.2018.11.025.

Chahat Raj, Anjishnu Mukherjee, Aylin Caliskan, Antonios Anastasopoulos, and Ziwei Zhu. Breaking bias, building bridges: Evaluation and mitigation of social biases in LLMs via contact hypothesis. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, volume 7, pages 1180–1189, 2024. doi:10.1609/aies.v7i1.31715. URL https://ojs.aaai.org/index.php/AIES/article/view/31715.

Peter Romero, Stephen Fitz, and Teruo Nakatsuma. Do GPT language models suffer from split personality disorder? the advent of substrate-free psychometrics, 2024. URL https://arxiv.org/abs/2408.07377.

Scott R. Ross, Javier Moltó, Rosario Poy, Pilar Segarra, M. Carmen Pastor, and Susana Montañés. Gray's model and psychopathy: BIS but not BAS differentiates primary from secondary psychopathy in noninstitutionalized young adults. *Personality and Individual Differences*, 43(7):1644–1655, 2007. doi:10.1016/j.paid.2007.04.020.

Bronson Schoen, Evgenia Nitishinskaya, Mikita Balesni, et al. Stress testing deliberative alignment for anti-scheming training, 2025. URL https://arxiv.org/abs/2509.15541.

Gregory Serapio-García, Mustafa Safdari, Clément Crepy, Luning Sun, Stephen Fitz, Peter Romero, Marwa Abdulhai, Aleksandra Faust, and Maja Matarić. A psychometric framework for evaluating and shaping personality traits in large language models. *Nature Machine Intelligence*, 7(12):1954–1968, 2025. doi:10.1038/s42256-025-01115-6. URL https://www.nature.com/articles/s42256-025-01115-6.

Joar Skalse, Nikolaus H. R. Howe, Dmitrii Krasheninnikov, and David Krueger. Defining and characterizing reward hacking. In *Advances in Neural Information Processing Systems*, volume 35, pages 9460–9471, 2022. doi:10.52202/068431-0687. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/3d719fee332caa23d5038b8a90e81796-Abstract-Conference.html.

Jennifer L. Skeem, Devon L. L. Polaschek, Christopher J. Patrick, and Scott O. Lilienfeld. Psychopathic personality: Bridging the gap between scientific evidence and public policy. *Psychological Science in the Public Interest*, 12(3): 95–162, 2011. doi:10.1177/1529100611426706.

Sarah Francis Smith and Scott O. Lilienfeld. The response modulation hypothesis of psychopathy: A meta-analytic and narrative analysis. *Psychological Bulletin*, 141(6):1145–1177, 2015. doi:10.1037/bul0000024.

James W. A. Strachan, Dalila Albergo, Giulia Borghini, Oriana Pansardi, Eugenio Scaliti, Saurabh Gupta, Krati Saxena, Alessandro Rufo, Stefano Panzeri, Guido Manzi, Michael S. A. Graziano, and Cristina Becchio. Testing theory of mind in large language models and humans. *Nature Human Behaviour*, 8(7):1285–1295, 2024. doi:10.1038/s41562-024-01882-z. URL https://www.nature.com/articles/s41562-024-01882-z.

Elham Tabassi. Artificial intelligence risk management framework (AI RMF 1.0). Technical Report NIST AI 100-1, National Institute of Standards and Technology, 2023. URL https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-ai-rmf-10.

Alexander Matt Turner, Logan Smith, Rohin Shah, Andrew Critch, and Prasad Tadepalli. Optimal policies tend to seek power. In *Advances in Neural Information Processing Systems*, volume 34, pages 23063–23074. Curran Associates, Inc., 2021. URL https://proceedings.neurips.cc/paper/2021/hash/c26820b8a4c1b3c2aa868d6d57e14a79-Abstract.html.

Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. In *Advances in Neural Information Processing Systems*, volume 36, pages 74952–74965, 2023. doi:10.52202/075280-3275. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/ed3fea9033a80fea1376299fa7863f4a-Abstract-Conference.html.

Kim Peter Wabersich and Melanie N. Zeilinger. A predictive safety filter for learning-based control of constrained nonlinear dynamical systems. *Automatica*, 129:109597, 2021. doi:10.1016/j.automatica.2021.109597.

Jenny Zhang, Shengran Hu, Cong Lu, Robert Tjarko Lange, and Jeff Clune. Darwin Gödel Machine: Open-ended evolution of self-improving agents. In *The Fourteenth International Conference on Learning Representations*, 2026. URL https://openreview.net/forum?id=pUpzQZTvGY. Version 3, revised 12 March 2026.

Shichang Zhang, Tessa Han, Usha Bhalla, and Himabindu Lakkaraju. Towards unified attribution in explainable AI, data-centric AI, and mechanistic interpretability, 2025. URL https://arxiv.org/abs/2501.18887.